



V93000 Specifications

V93000 Wave Scale Millimeter

Version 8.6.0

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Notices

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Wave Scale Millimeter specifications

The Wave Scale Millimeter (WSMM) solution is an extension of the V93000 DUT interface providing an integrated test solution for 5G and WiGig applications. The solution is based on the Wave Scale RF base card (WSRF-6 or WSRF-8) and extends its capabilities for IF (5.85 to 18 GHz) using RFIM-18 modules, and for millimeter wave frequencies using Remote Mixing Modules (RMMs).

Three versions of RMMs are supported: RMM-44 (operational limit 24 to 30 GHz, 35 to 44 GHz) and RMM-53 (operational limit 24 to 52.8 GHz) for 5G applications, and RMM-72 (operational limit 57 to 70 GHz) for WiGig applications.

Terms and conditions

spec	Specifications describe warranted product performance.
typ	Characteristics are included as typical values describe a performance that is not covered by the product warranty. They describe a performance that is met by 80 % of the units with a 95 % confidence level.
meas	Measured values characterize expected product performance by means of measurements provided on individual samples.

- Note

All Wave Scale Millimeter (WSMM) specifications are including Advantest's 1.85 mm DUT board connector, providing a standardized coaxial interface.
- Note

The Wave Scale Millimeter (WSMM) specifications are only valid with LO source WSPS .

WSMM stimulus

WSMM stimulus accuracy

Stimulus accuracy conditions and notes:

- All values are typical according to *Terms and conditions* on page 4, unless noted.
- All power levels are CW signals in a 50 Ω system, unless noted.
- All values are valid after the adjustment. For a new product, 2 weeks of settling time are required until the performance is final. Within this time, the accuracy can settle up to 1.2 dB.
- Non-default parameters in SmarTest 8:
 - optimization = balanced
 - optimizationNoiseBandwidth = 1e-100 Hz

Note All values are in dB.

RMM-44 without Switch

Typical

Frequency range (GHz)	Power level range (dBm)	
	$-80 \leq L_v < -50$	$-50 \leq L_v \leq +4$
$24 \leq f \leq 30$	± 1.27	± 0.69
$35 \leq f \leq 42$	± 1.14	± 0.90
$42 < f \leq 44$	± 1.89	± 1.59 ^[1]

RMM-44 with 1:8 Switch

Typical

Frequency range (GHz)	Power level range (dBm)		
	$-80 \leq L_v < -50$	$-50 \leq L_v \leq -2$	$-2 < L_v \leq +2$
$24 \leq f \leq 30$	± 0.98	± 0.47	± 0.90
$35 \leq f \leq 40$	± 1.09	± 0.63	± 0.81 ^[2]

RMM-53 without switch

Typical

Frequency range (GHz)	Power level range (dBm)		
	$-80 \leq L_v < -60$	$-60 \leq L_v < -10$	$-10 \leq L_v \leq +4$
$24 \leq f \leq 27$	± 0.95	± 0.46	± 0.49
$27 < f \leq 35$	± 0.89 ^[3]	± 0.53	± 0.52
$35 < f \leq 50$	± 0.95	± 0.60	± 0.58

^[1] for $-50 \leq L_v \leq +2$

^[2] for $-2 < L_v \leq 0$

^[3] for $-70 \leq L_v < -60$

Frequency range (GHz)	Power level range (dBm)		
	$-80 \leq L_v < -60$	$-60 \leq L_v < -10$	$-10 \leq L_v \leq +4$
$50 < f \leq 52.8$	± 0.43 [4]	± 0.44 [4]	± 0.67 [4]

RMM-53 with 1:8 switch

Typical

Frequency range (GHz)	Power level range (dBm)			
	$-80 \leq L_v < -60$	$-60 \leq L_v < -10$	$-10 \leq L_v < -4$	$-4 \leq L_v \leq +2$
$24 \leq f \leq 27$	± 0.58	± 0.44	± 0.46	± 0.59
$27 < f \leq 31$	± 0.69 [3]	± 0.52	± 0.50	± 0.60
$31 < f \leq 42$	± 0.69	± 0.54	± 0.55	± 0.75 [5]
$42 < f \leq 50$	± 0.72	± 0.60	± 0.64	± 0.99 [6]
$50 < f \leq 52.8$	± 0.45 [4]	± 0.43 [4]	± 0.74 [4][7]	± 0.76 [4][6][8]

RMM-72 without Switch

Typical

Frequency range (GHz)	Power level range (dBm)		
	$-65 \leq L_v < -60$	$-60 \leq L_v < -50$	$-50 \leq L_v \leq +6$
$57 \leq f \leq 67$	± 1.07	± 0.73	± 0.58

RMM-72 with 1:8 Switch

Typical

Frequency range (GHz)	Power level range (dBm)		
	$-65 \leq L_v < -60$	$-60 \leq L_v \leq -1$	$-1 < L_v \leq +1$
$57 \leq f \leq 67$	± 0.81	± 0.68	± 1.18

WSMM stimulus noise floor

Stimulus noise floor conditions and notes:

- All values are typical according to *Terms and conditions* on page 4, unless noted.
- Non-default parameters in SmarTest 8:
 - optimization = noiseFloor
- Stimulus CW frequency has an offset of 40 MHz^[9] against specified frequency.

RMM-44 without switch

Typical

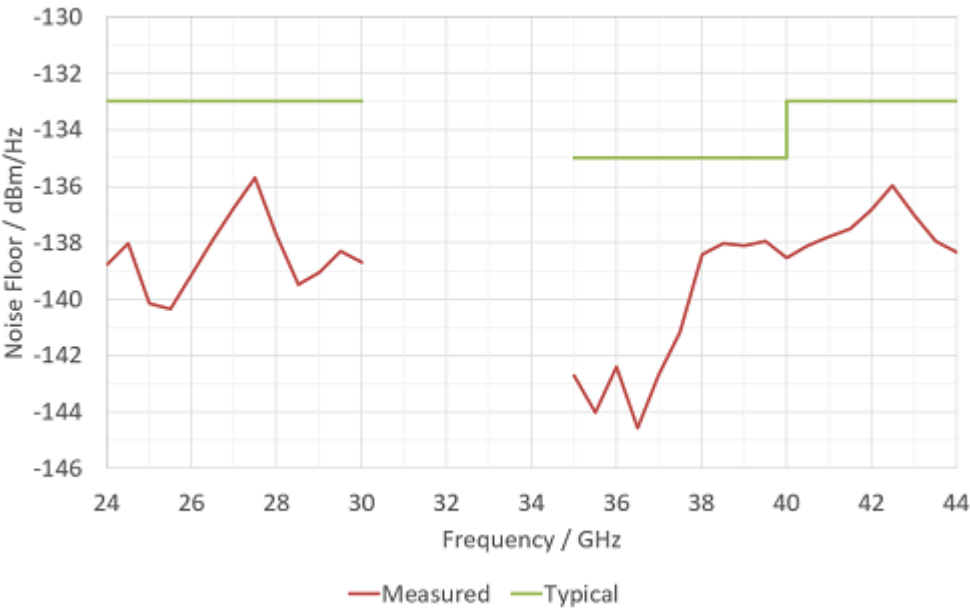
[4] Measured

[5] $-4 \leq L_v < -1$ for $f > 40$ GHz[6] $-4 \leq L_v < -2$ [7] $-10 \leq L_v < -7$ for $f \geq 52$ GHz[8] $50 < f \leq 50.5$ GHz

[9] offset was adjusted for some frequencies to avoid spurs

Carrier frequency range (GHz)	Noise floor (dBm/Hz)
$24 \leq f \leq 30$	-133
$35 \leq f \leq 40$	-135
$40 < f \leq 44$	-133

Measured with carrier power = 0 dBm



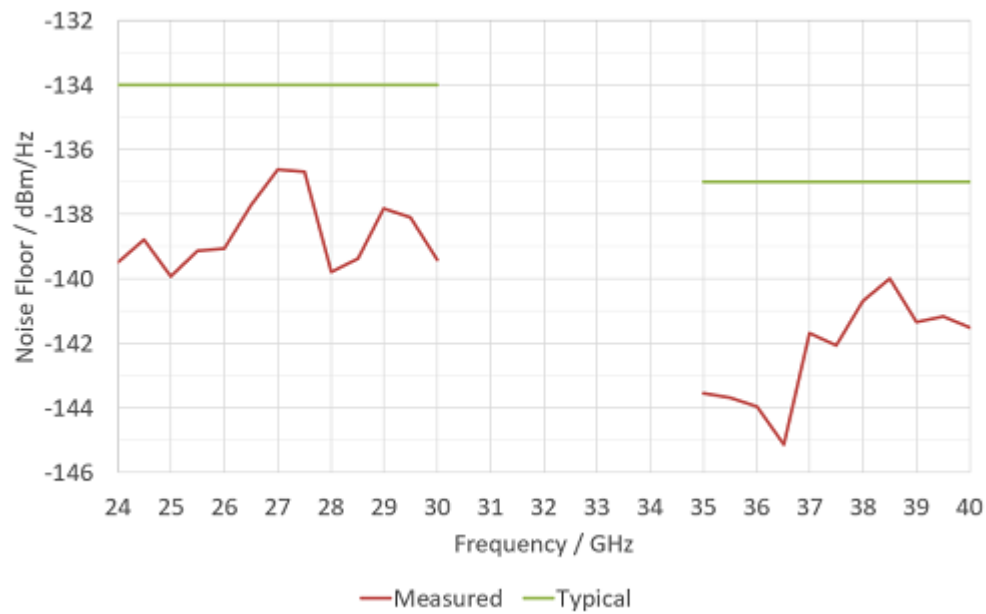
RMM-44 without switch

RMM-44 with 1:8 switch

Typical

Carrier frequency range (GHz)	Noise floor (dBm/Hz)
$24 \leq f \leq 30$	-134
$35 \leq f \leq 40$	-137

Measured with carrier power = -5 dBm



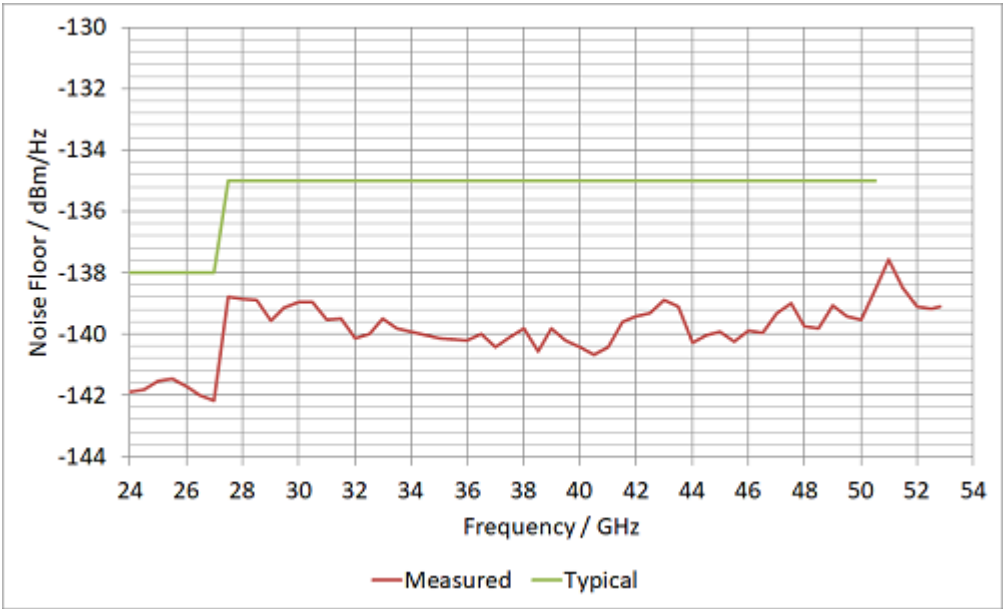
RMM-44 with 1:8 switch

RMM-53 without switch

Typical

Carrier frequency range (GHz)	Noise floor (dBm/Hz)
$24 \leq f \leq 27$	-138
$27 < f \leq 50$	-135
$52 < f \leq 52.8$	-135 ^[10]

Measured with carrier power = -2 dBm



RMM-53 without switch

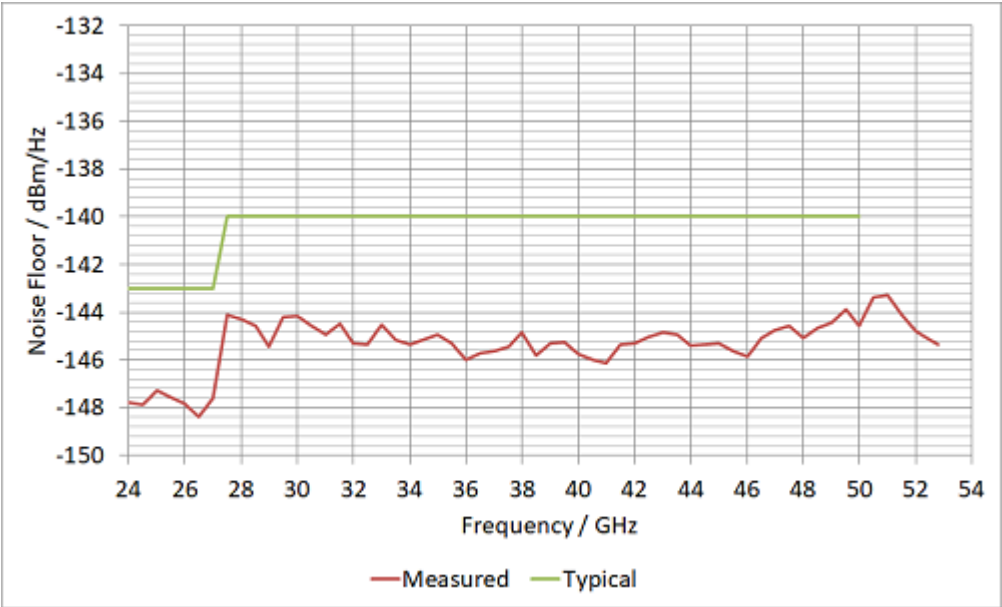
^[10] Measured

RMM-53 with 1:8 switch

Typical

Carrier frequency range (GHz)	Noise floor (dBm/Hz)
$24 \leq f \leq 27$	-143
$27 < f \leq 50$	-140
$50 < f \leq 52.8$	-140 ^[10]

Measured with carrier power = -7 dBm



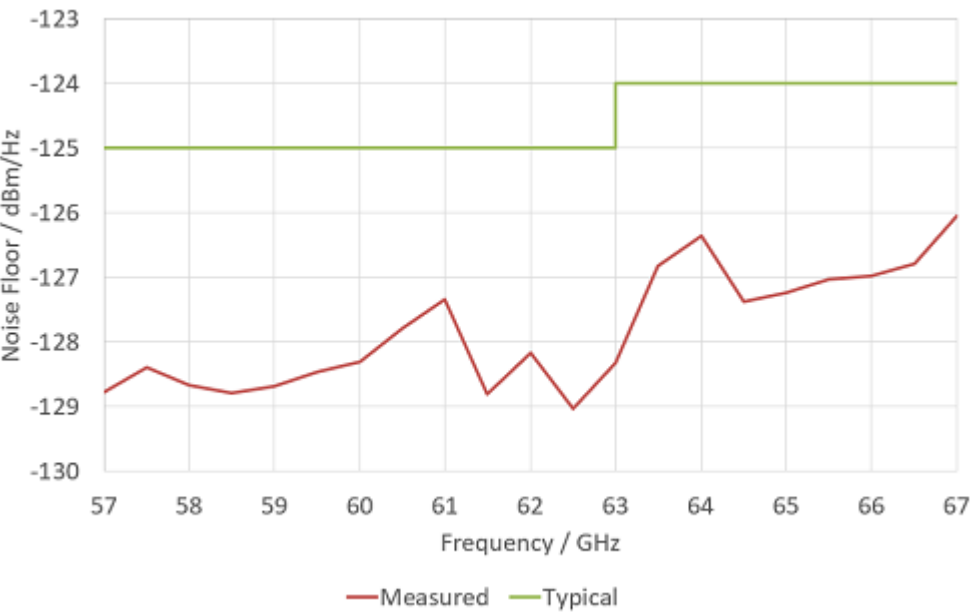
RMM-53 with 1:8 switch

RMM-72 without switch

Typical

Carrier frequency range (GHz)	Noise floor (dBm/Hz)
$57 \leq f \leq 63$	-125
$63 < f \leq 67$	-124

Measured with carrier power = -2 dBm



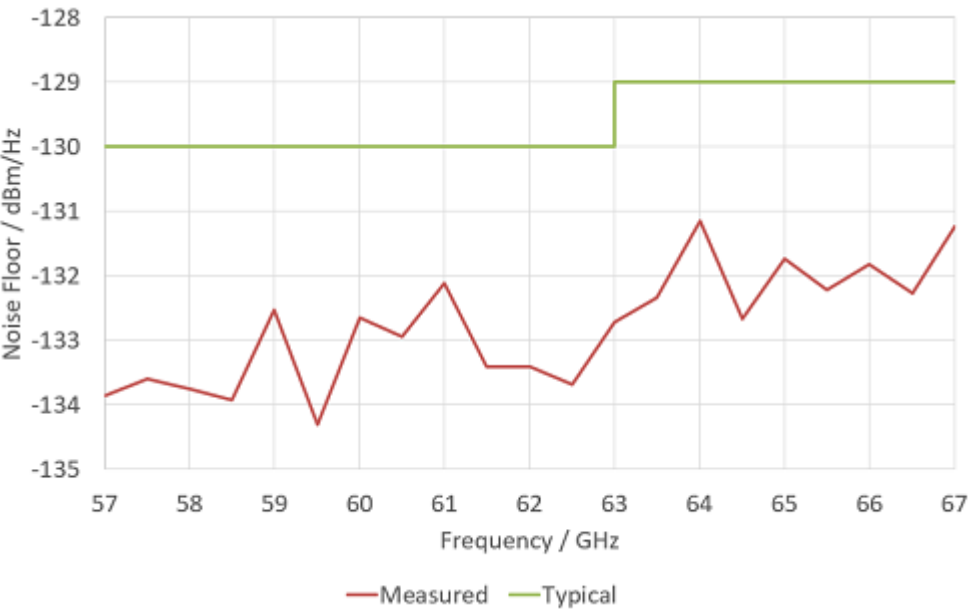
RMM-72 without switch

RMM-72 with 1:8 switch

Typical

Carrier frequency range (GHz)	Noise floor (dBm/Hz)
$57 \leq f \leq 63$	-130
$63 < f \leq 67$	-129

Measured with carrier power = -7 dBm



RMM-44 with 1:8 switch

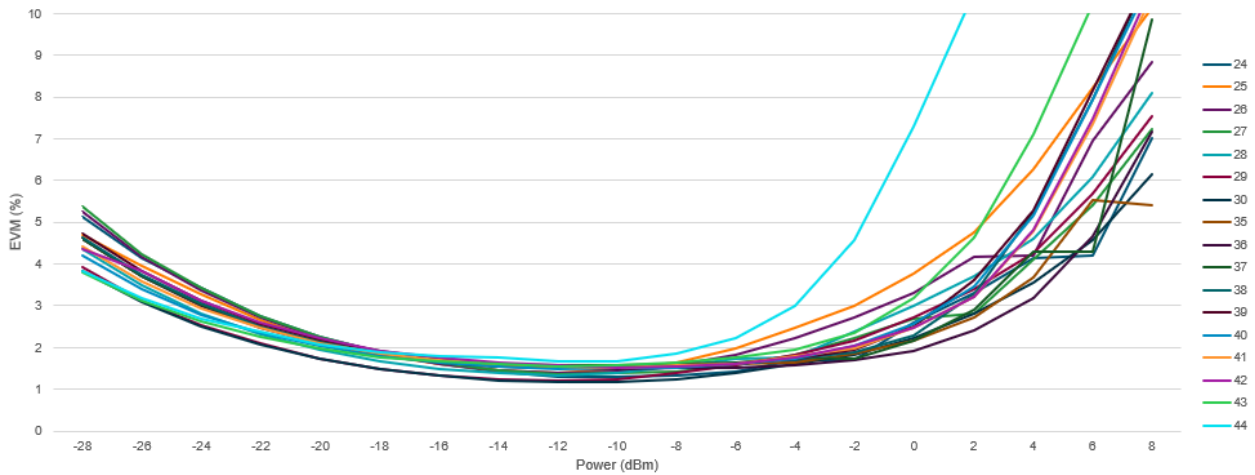
WSMM stimulus EVM for RMM-44

Stimulus EVM for RMM-44 conditions and notes:

- All values are meas values, according to *Terms and conditions* on page 4.

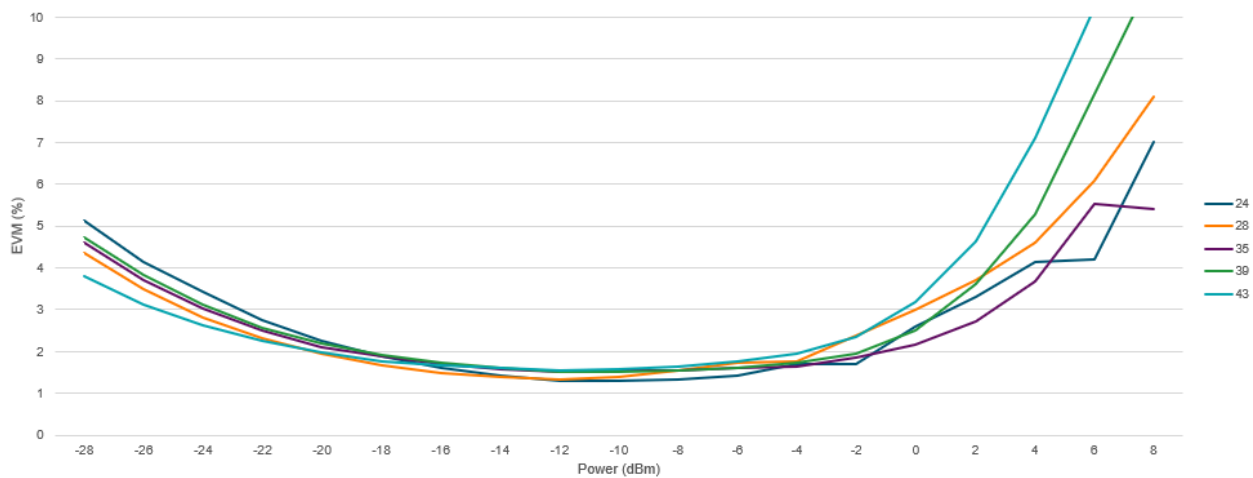
- Non-default parameters in SmarTest 8
 - optimization = balanced
 - optimizationNoiseBandwidth = 1e6
- Base Frequency left as default
- Waveform = NR5G_MCS27_UL_100MHz_64QAM

Stimulus EVM for RMM-44 without switch 24 GHz to 44 GHz

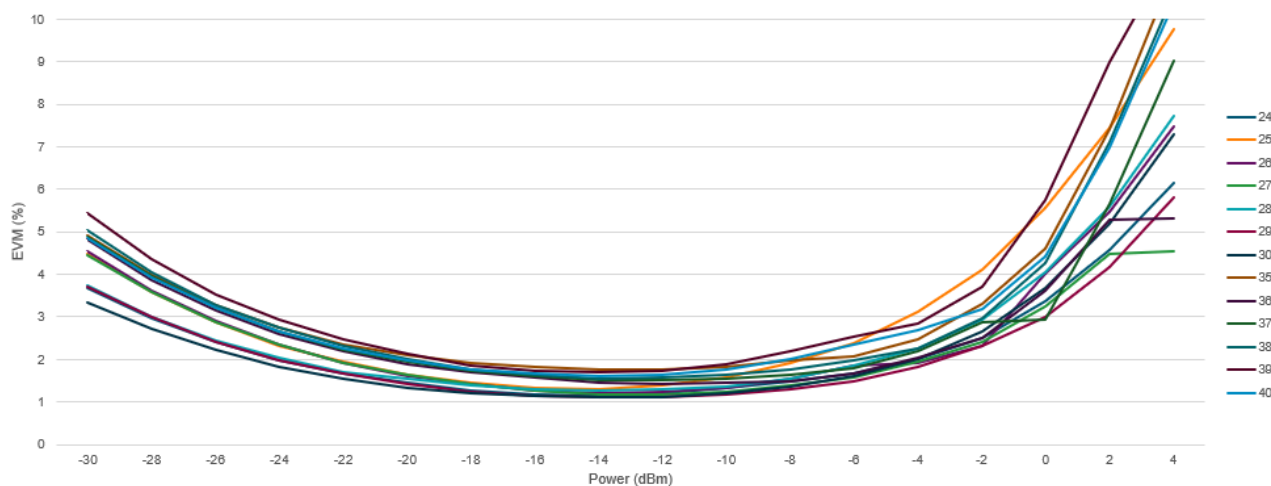
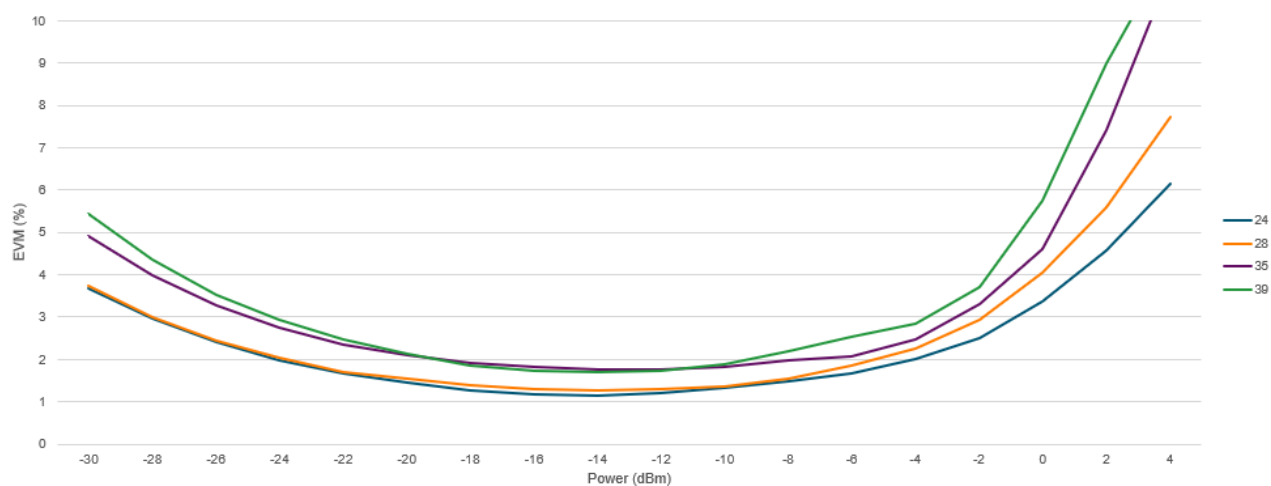


Stim EVM x1 RMM-44

Stimulus EVM RMM-44 without switch 24 GHz to 44 GHz with few frequencies

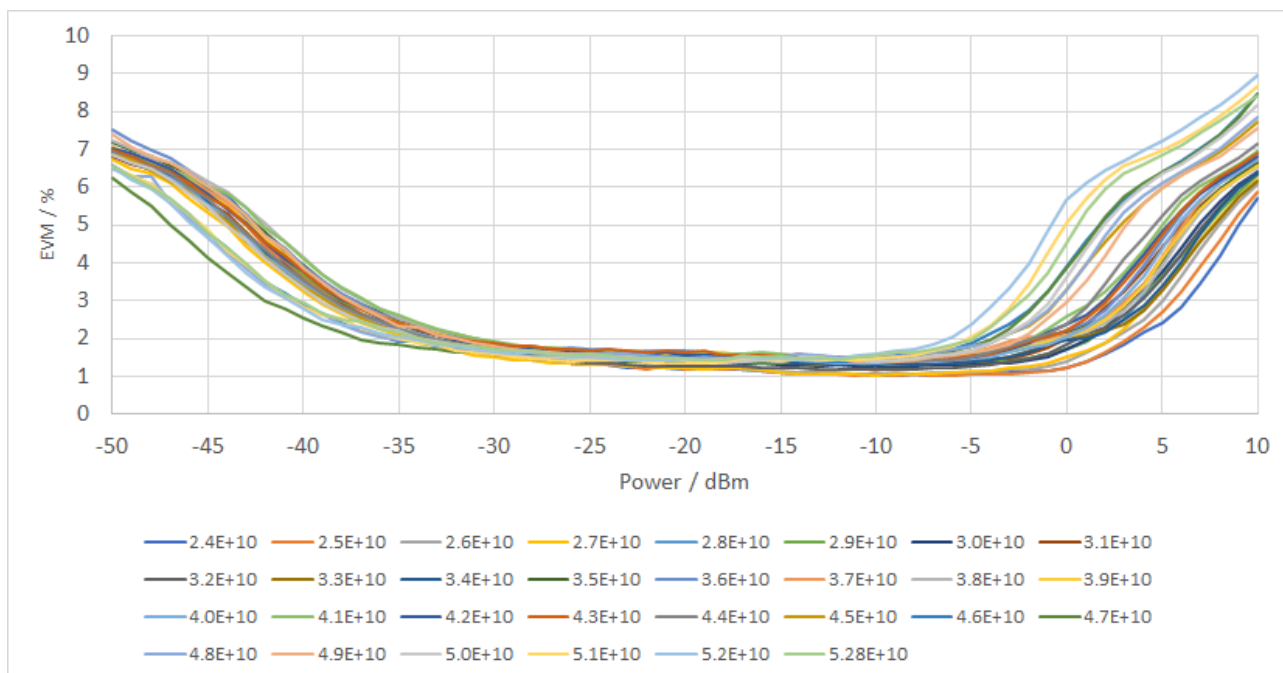


Stimulus EVM (x1) RMM-44 direct 24 GHz to 44 GHz with few frequencies

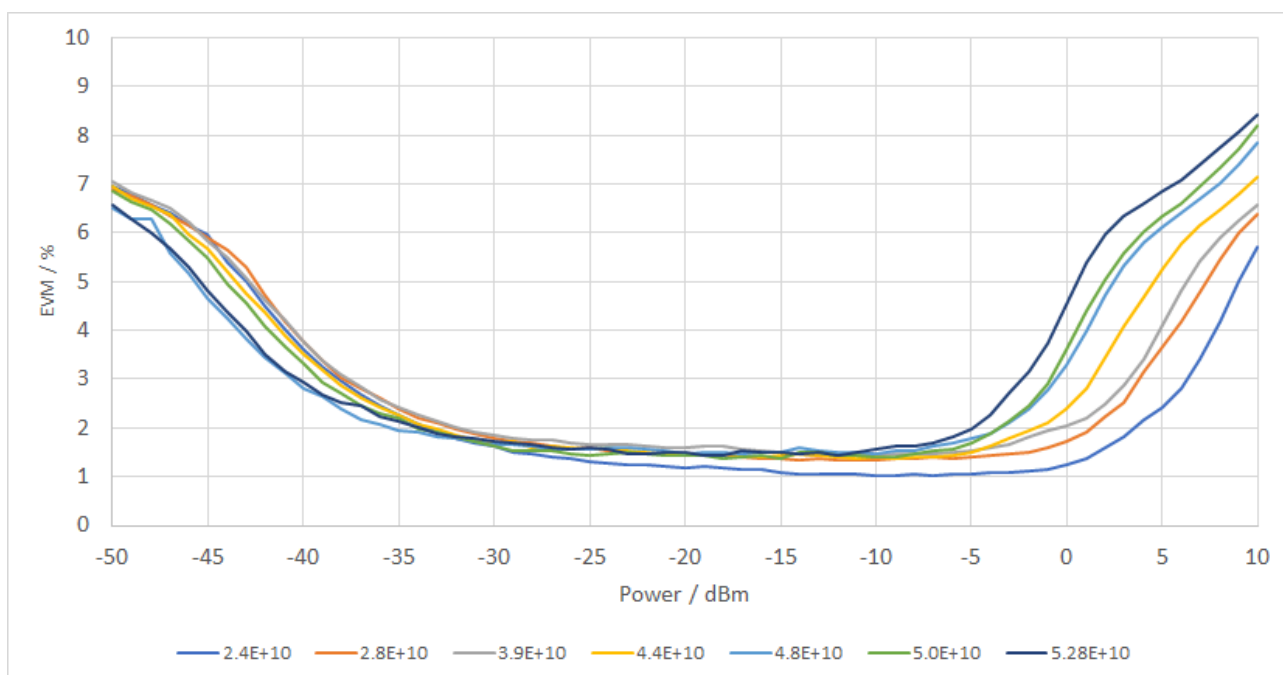
Stimulus EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz**Stimulus EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz****Stimulus EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz with few frequencies****Stimulus EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz with few frequencies****WSMM stimulus EVM for RMM-53**

Stimulus EVM for RMM-53 conditions and notes:

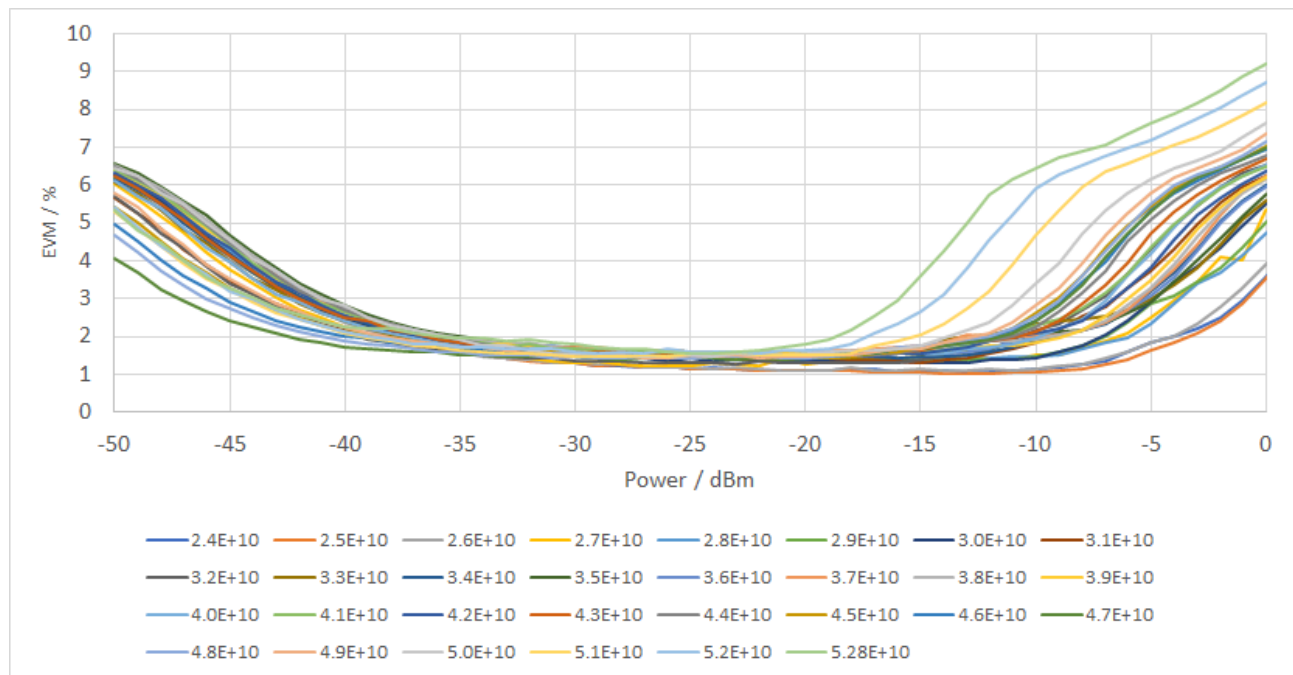
- All values are meas values, according to *Terms and conditions* on page 4.
- Non-default parameters in SmarTest 8
 - optimization = balanced
 - optimizationNoiseBandwidth = 10e6
 - Base Frequency = 3.9e9
- Waveform = UL_MCS27_256QAM_100MHz_FR2_122p88Msps

Stimulus EVM RMM-53 without switch 24 GHz to 52.8 GHz

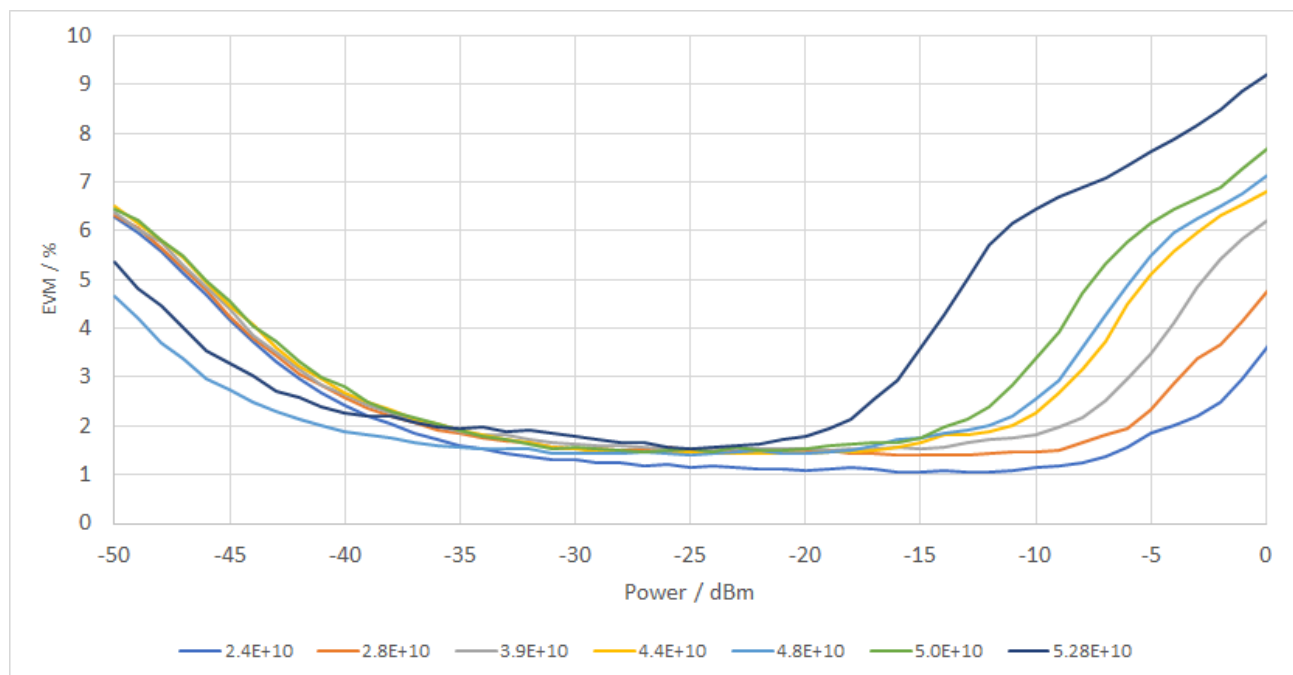
Stim EVM RMM53 without switch

Stimulus EVM RMM-53 without switch 24 GHz to 52.8 GHz with few frequencies

Stim EVM RMM53 without switch few freq

Stimulus EVM RMM-53 with 1:8 switch 24 GHz to 52.8 GHz

Stim EVM RMM53 with switch

Stimulus EVM RMM-53 with 1:8 switch 24 GHz to 52.8 GHz with few frequencies

Stim EVM Rmm53 with switch few freq

WSMM stimulus maximum output power

This is a measure of guaranteed maximum output power, measured as saturation power.

It is measured with the following conditions:

- All values are meas values, according to *Terms and conditions* on page 4.

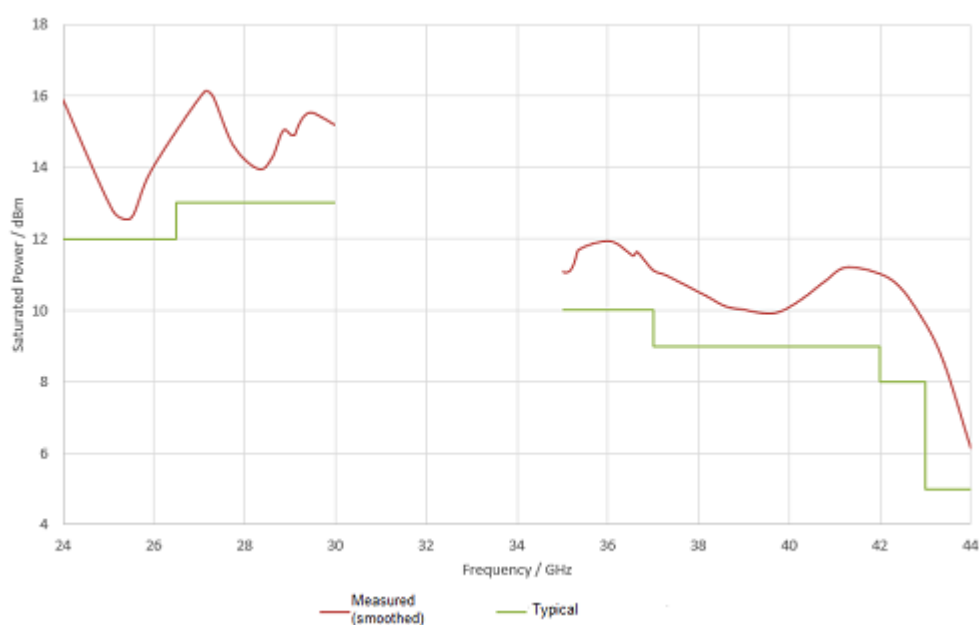
- Non-default parameters in SmarTest 8
 - optimization = balanced
 - optimizationNoiseBandwidth = 1e-100 Hz

RMM-44 without Switch

Typical

power setting = +30 dBm

Carrier frequency range (GHz)	Saturated power (dBm)
$24 \leq f < 26.5$	+12
$26.5 \leq f \leq 30$	+13
$35 \leq f \leq 37$	+10
$37 < f \leq 42$	+9
$42 < f \leq 43$	+8
$43 < f \leq 44$	+5



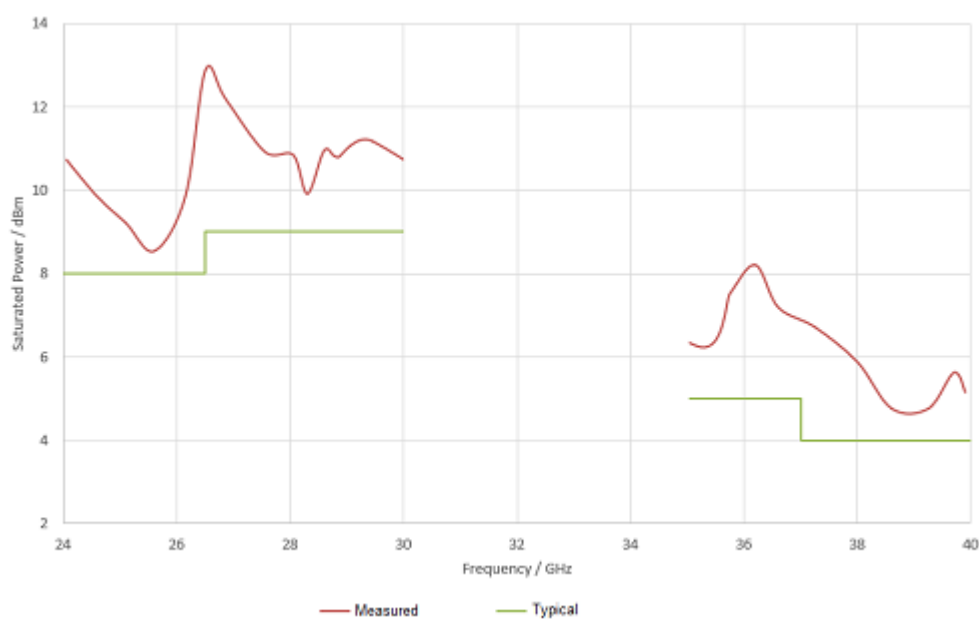
RMM-44 without switch

RMM-44 with 1:8 switch

Typical

power setting = +30 dBm

Carrier frequency range (GHz)	Saturated power (dBm)
$24 \leq f < 26.5$	+8
$26.5 \leq f \leq 30$	+9
$35 \leq f \leq 37$	+5
$37 < f \leq 40$	+4



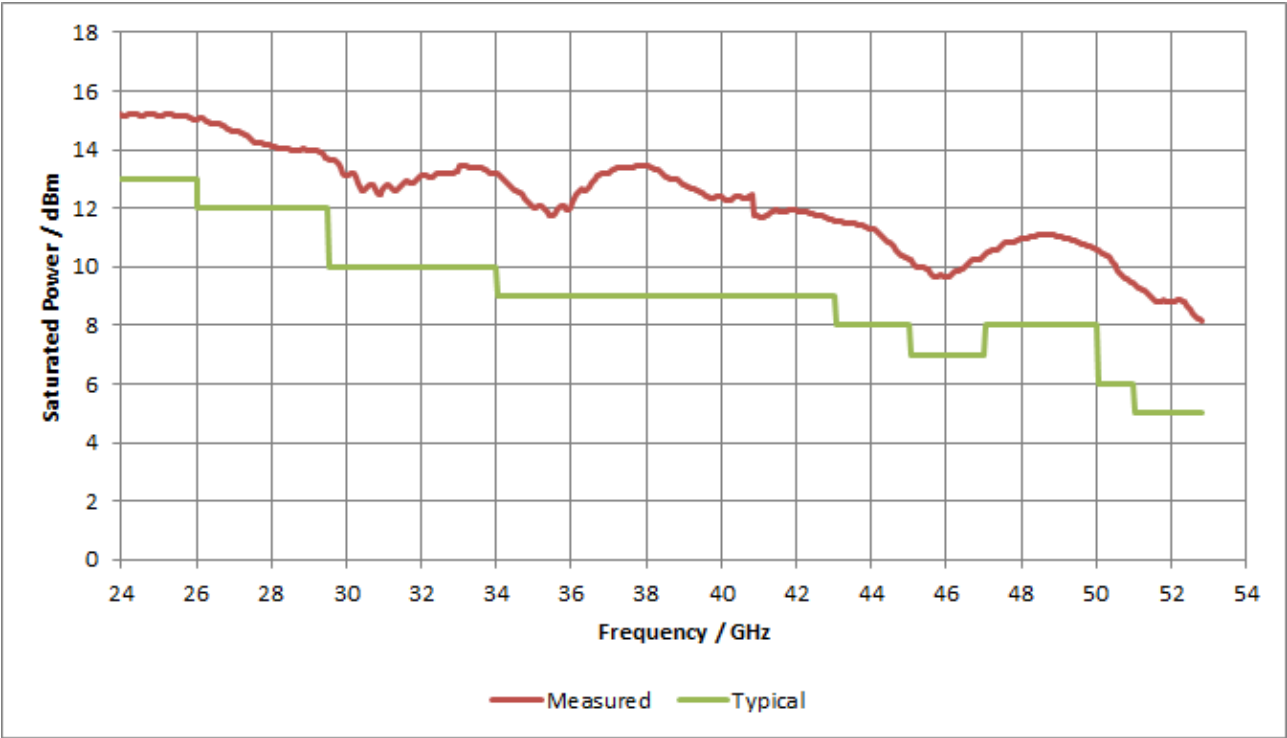
RMM-44 with 1:8 switch

RMM-53 without switch

Typical

Power setting = +20 dBm

Frequency (GHz)	Saturated power (dBm)
$24 \leq f \leq 26$	+13
$26.0 < f \leq 29.5$	+12
$29.5 < f \leq 34$	+10
$34 < f \leq 43$	+9
$43 < f \leq 45$	+8
$45 < f \leq 47$	+7
$47 < f \leq 50$	+8
$50 < f \leq 51$	+6
$51 < f \leq 52.8$	+5



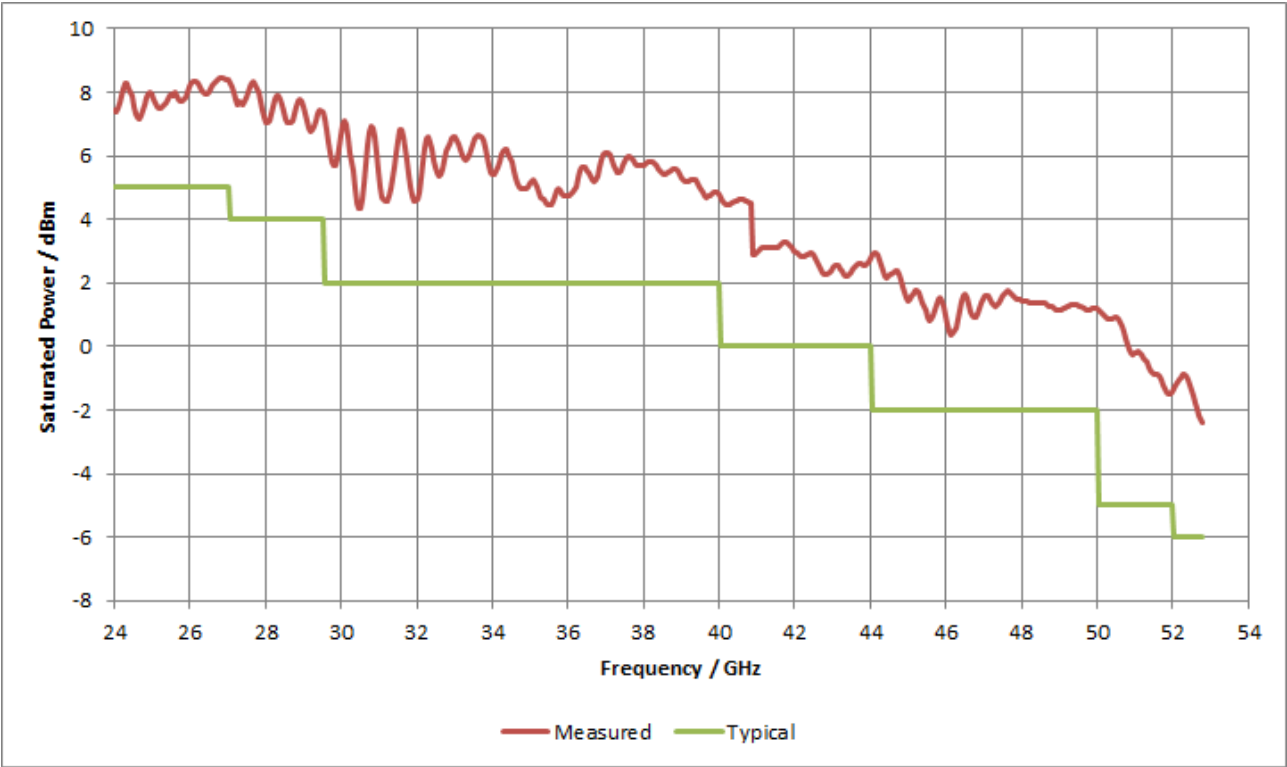
Stim power saturated RMM-53 without switch

RMM-53 with 1:8 switch

Typical

Power setting = +20 dBm

Frequency (GHz)	Saturated power (dBm)
$24 \leq f \leq 27$	+5
$27 < f \leq 29.5$	+4
$29.5 < f \leq 40$	+2
$40 < f \leq 44$	0
$44 < f \leq 50$	-2
$50 < f \leq 52$	-5
$52 < f \leq 52.8$	-6



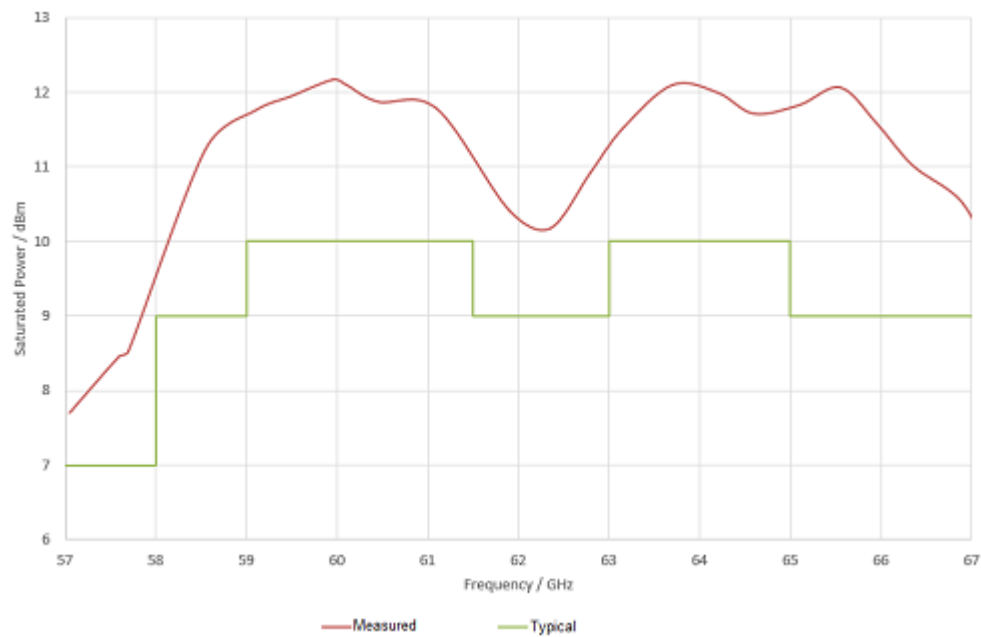
Stim power saturated RMM-53 with switch

RMM-72 without switch

Typical

power setting = +30 dBm

Carrier frequency range (GHz)	Saturated power (dBm)
$57 \leq f < 58$	+7
$58 \leq f < 59$	+9
$59 \leq f \leq 61.5$	+10
$61.5 < f < 63$	+9
$63 \leq f \leq 65$	+10
$65 < f \leq 67$	+9



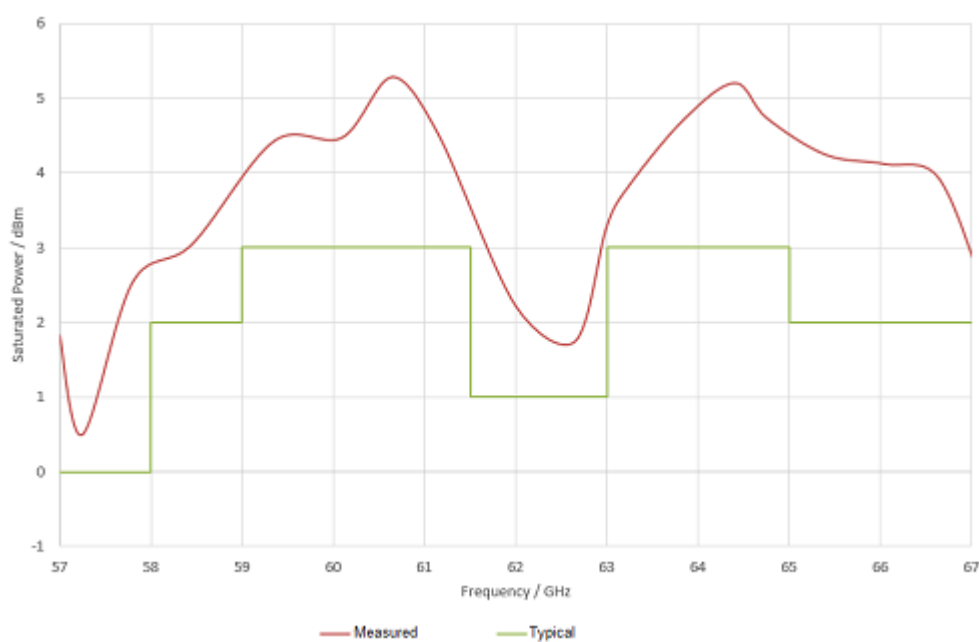
RMM-72 without switch

RMM-72 with 1:8 switch

Typical

power setting = +30 dBm

Carrier frequency range (GHz)	Saturated power (dBm)
$57 \leq f < 58$	+0
$58 \leq f < 59$	+2
$59 \leq f \leq 61.5$	+3
$61.5 < f < 63$	+1
$63 \leq f \leq 65$	+3
$65 < f \leq 67$	+2



RMM-44 with 1:8 switch

WSMM stimulus phase noise

Stimulus phase noise conditions and notes:

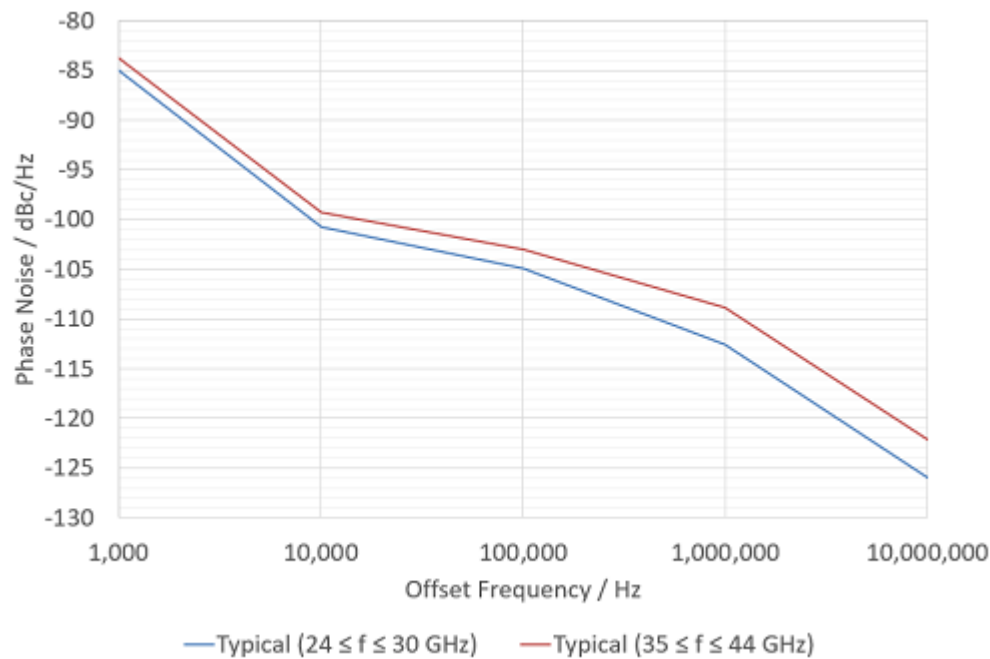
- All values are typical according to *Terms and conditions* on page 4, unless noted.
- Non-default parameters in SmarTest 8:
 - optimization = noiseFloor
 - optimizationNoiseBandwidth = 80 MHz
- Phase noise was measured with a -3 dBm carrier signal.
- Spurs are not taken into consideration.

Note All values are in dBc/Hz.

RMM-44 without switch

Typical

Carrier frequency range (GHz)	Offset Frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
$24 \leq f \leq 30$	-85.0	-100.8	-104.9	-112.6	-126.0
$35 \leq f \leq 44$	-83.7	-99.3	-103.0	-108.9	-122.2

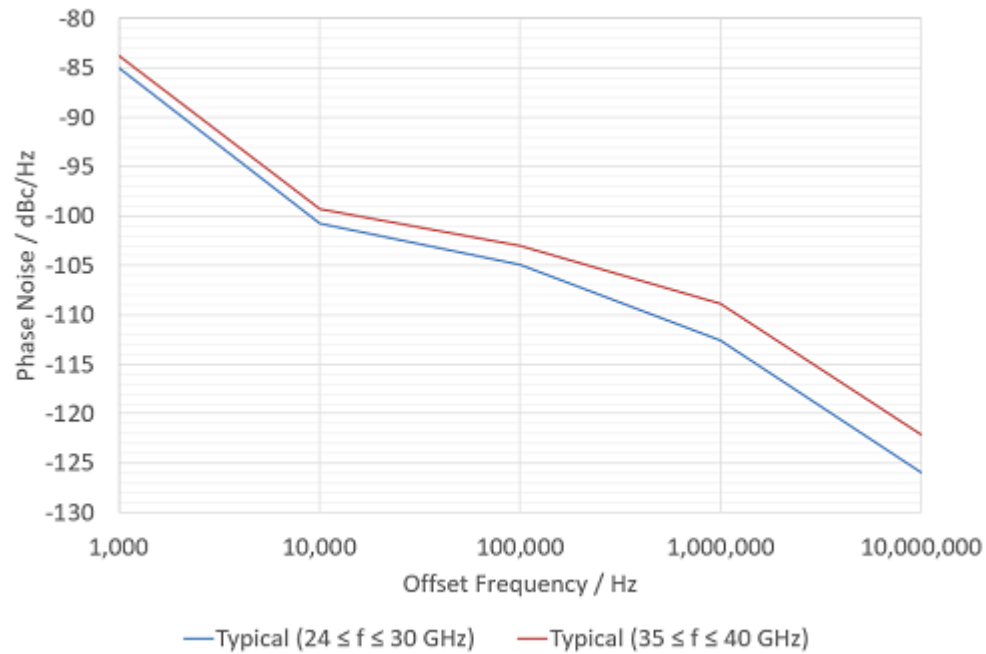


RMM-44 without switch

RMM-44 with 1:8 switch

Typical

Carrier frequency range (GHz)	Offset Frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
24 ≤ f ≤ 30	-85.0	-100.8	-104.9	-112.6	-126.0
35 ≤ f ≤ 40	-83.7	-99.3	-103.0	-108.9	-122.2

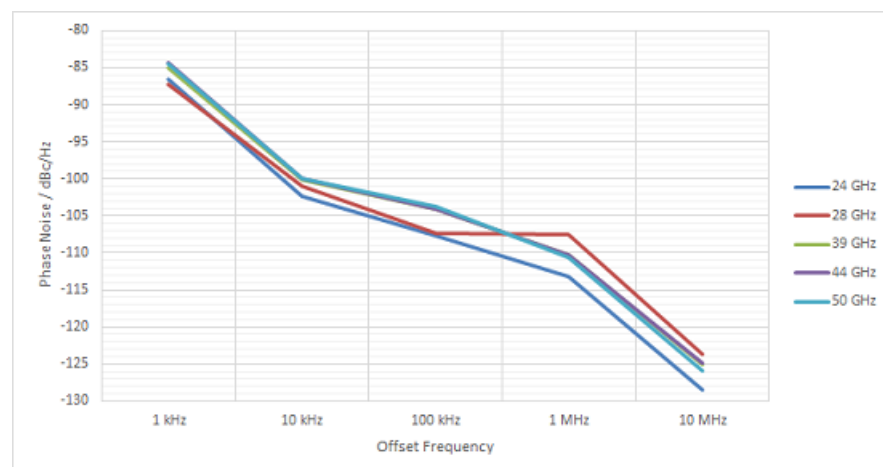


RMM-44 with 1:8 switch

RMM-53 without switch or with 1:8 switch

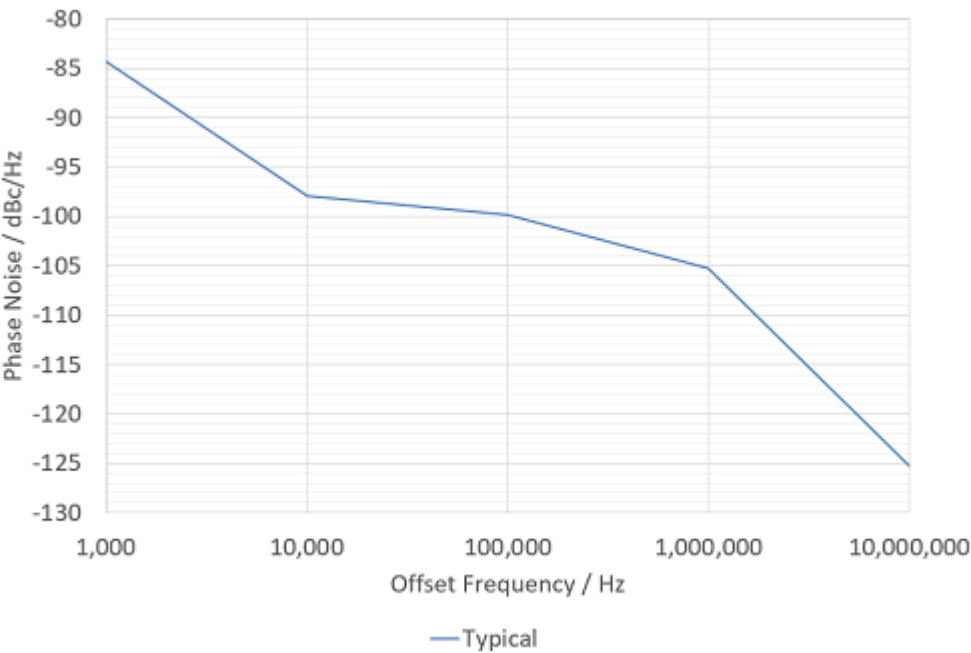
Typical

Carrier frequency range (GHz)	Offset Frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
$24 \leq f \leq 27$	-85.0	-101.1	-105.5	-112.1	-127.3
$27 < f \leq 32$	-82.4	-99.5	-105.5	-106.4	-122.7
$32 < f \leq 44$	-82.4	-98.6	-102.3	-108.4	-123.1
$44 < f \leq 50$	-82.4	-98.6	-101.5	-108.4	-124

**RMM-53 with 1:8 switch****RMM-72 without switch or with 1:8 switch**

Typical

Carrier frequency range (GHz)	Offset Frequency				
	1 kHz	10 kHz	100 kHz	1 MHz	10 MHz
$57 \leq f \leq 67$	-84.3	-97.9	-99.9	-105.2	-125.3



RMM-72 without switch

WSMM measure

WSMM measure accuracy

Measure accuracy conditions and notes:

- All values are typical according to *Terms and conditions* on page 4, unless noted.
- All power levels are CW signals, unless noted.
- Non-default parameters in SmarTest 8:
 - bandwidthOfInterest = 1 kHz
 - resultAveraging = 16

Note All values are in dB.

RMM-44 without Switch

Typical

mixReceiverPath = direct

Frequency range (GHz)	Power level range (dBm)		
	$-85 \leq L_v < -60$	$-60 \leq L_v < -5$	$-5 \leq L_v \leq +4$
$24 \leq f < 26$	± 0.99	± 0.94	± 0.81
$26 \leq f \leq 30$	± 0.71	± 0.65	± 0.56
$35 \leq f < 40$	± 1.08	± 0.92	± 0.81
$40 \leq f \leq 44$	± 0.93	± 0.77	± 0.68

The given values are valid after the adjustment. However, there will be a drift up to the value shown in the column *Monthly* of the next table per month over time. In case any brand new component (such as a new WSRF-6 card, a new RFIM-18, a new RMM and so on) is used, there will be an additional drift up to the value shown in the column *Initial* within the first two weeks. Afterwards, the initial drift will be gone and only the monthly drift will be taken into account.

Frequency range (GHz)	Additional Drift	
	Initial (dB)	Monthly (dB)
$24 \leq f < 26$	-	± 0.07
$26 \leq f \leq 30$	-	± 0.07
$35 \leq f < 40$	-	± 0.09
$40 \leq f \leq 44$	-	± 0.12

RMM-44 with 1:8 Switch

Typical

mixReceiverPath = direct

Frequency range (GHz)	Power level range (dBm)		
	$-85 \leq L_v < -60$	$-60 \leq L_v < -5$	$-5 \leq L_v \leq +4$
$24 \leq f < 26.5$	± 1.90	± 1.87	± 1.81
$26.5 \leq f \leq 30$	± 1.12	± 1.09	± 1.05

Frequency range (GHz)	Power level range (dBm)		
	$-85 \leq Lv < -60$	$-60 \leq Lv < -5$	$-5 \leq Lv \leq +4$
$35 \leq f \leq 40$	± 1.27	± 1.15	± 1.06

The given values are valid after the adjustment. However, there will be a drift up to the value shown in the column *Monthly* of the next table per month over time. In case any brand new component (such as a new WSRF-6 card, a new RFIM-18, a new RMM and so on) is used, there will be an additional drift up to the value shown in the column *Initial* within the first two weeks. Afterwards, the initial drift will be gone and only the monthly drift will be taken into account.

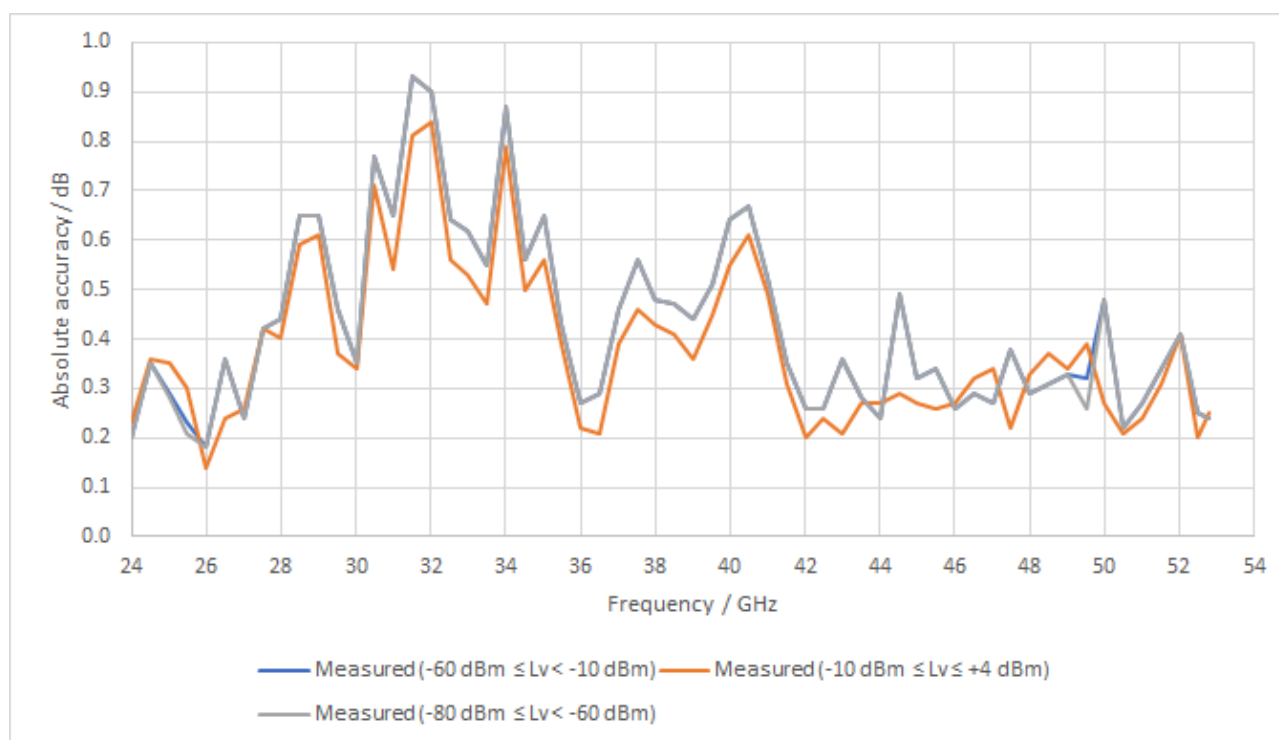
Frequency range (GHz)	Additional Drift	
	Initial (dB)	Monthly (dB)
$24 \leq f < 26.5$	-	± 0.07
$26.5 \leq f \leq 30$	-	± 0.07
$35 \leq f \leq 40$	-	± 0.17

RMM-53 without switch

Typical

Frequency Range (GHz)	Power level range (dBm)		
	$-80 \leq Lv < -60$	$-60 \leq Lv < -10$	$-10 \leq Lv \leq +4$
$24 \leq f \leq 25$	± 0.73	± 0.71	± 0.75
$25 < f \leq 26$	± 0.53	± 0.65	± 0.66
$26 < f \leq 30$	± 0.86	± 0.98	± 0.92
$30 < f \leq 36$	± 1.12	± 1.22	± 1.11
$36 < f \leq 40$	± 0.67	± 0.76	± 0.66
$40 < f \leq 42$	± 0.77	± 0.88	± 0.81
$42 < f \leq 50$	± 0.86	± 0.74	± 0.61
$50 < f \leq 52.8$	± 0.38 ^[11]	± 0.41 ^[11]	± 0.41 ^[11]

^[11] Measured



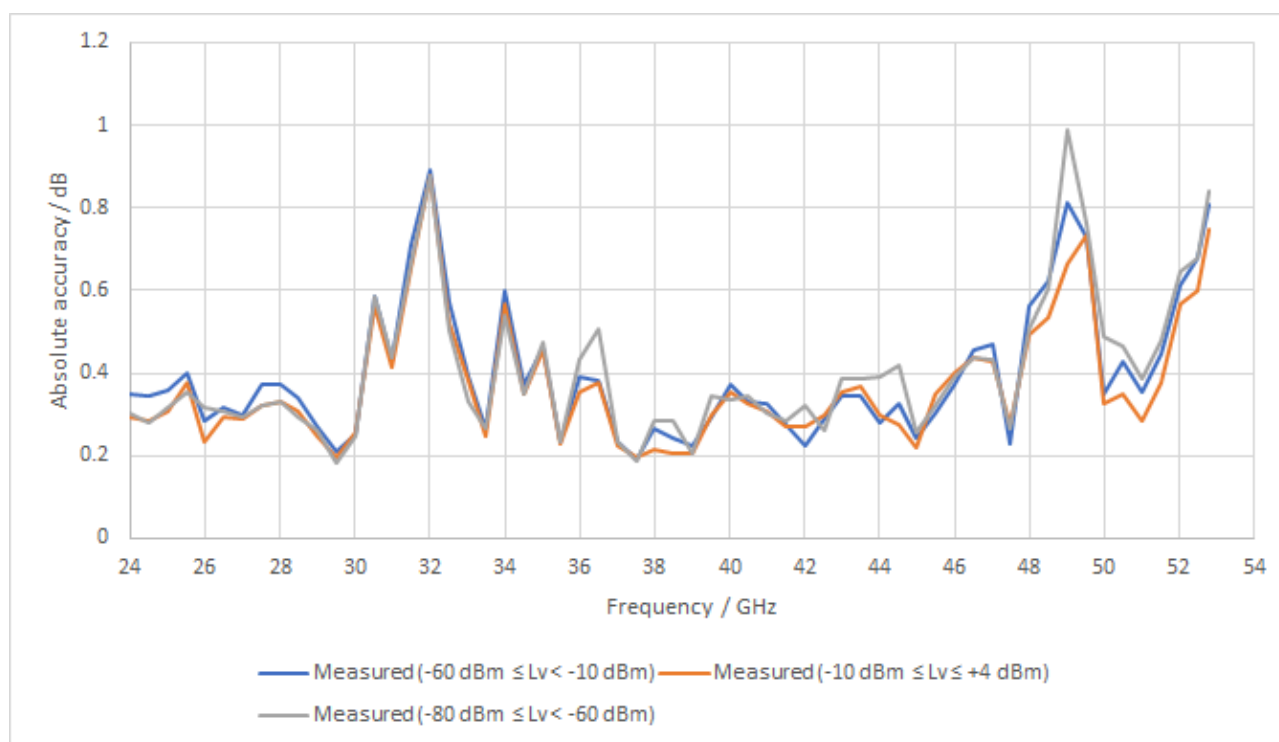
Meas accuracy RMM53 without switch

The given values are valid after the adjustment. For a new product, 2 weeks of settling time are required until the performance is final. Within this time, the accuracy can settle up to 1.2 dB.

RMM-53 with 1:8 switch

Typical

Frequency Range	Power level range (dBm)		
	$-80 \leq Lv < -60$	$-60 \leq Lv < -10$	$-10 \leq Lv \leq +4$
$24 \leq f \leq 25$	± 0.75	± 0.80	± 0.73
$25 < f \leq 30$	± 0.67	± 0.71	± 0.68
$30 < f \leq 35$	± 1.42	± 1.44	± 1.43
$35 < f \leq 37$	± 0.97	± 0.87	± 0.86
$37 < f \leq 45$	± 0.82	± 0.71	± 0.70
$45 < f \leq 50$	± 1.37	± 1.13	± 1.03
$50 < f \leq 52.8$	± 0.85 ^[11]	± 0.81 ^[11]	± 0.75 ^[11]



Meas accuracy RMM53 with switch

The given values are valid after the adjustment. For a new product, 2 weeks of settling time are required until the performance is final. Within this time, the accuracy can settle up to 1.2 dB.

RMM-72 without Switch

Typical

mixReceiverPath = direct

Frequency range (GHz)	Power level range (dBm)	
	-85 ≤ Lv < -60	-60 ≤ Lv ≤ +5
57 ≤ f < 59.5	±1.58	±1.41
59.5 ≤ f ≤ 61.5	±1.42	±1.26
61.5 < f < 63	±2.50	±2.57
63 ≤ f ≤ 65	±1.61	±1.42 ^[12]
65 < f ≤ 67	±1.97	±1.75 ^[13]

The given values are valid after the adjustment. However, there will be a drift up to the value shown in the column *Monthly* of the next table per month over time. In case any brand new component (such as a new WSRF-6 card, a new RFIM-18, a new RMM and so on) is used, there will be an additional drift up to the value shown in the column *Initial* within the first two weeks. Afterwards, the initial drift will be gone and only the monthly drift will be taken into account.

^[12] for -60 ≤ Lv ≤ +4 dBm

^[13] for -60 ≤ Lv ≤ +3 dBm

Frequency range (GHz)	Additional Drift	
	Initial (dB)	Monthly (dB)
$57 \leq f < 59.5$	-	± 0.07
$59.5 \leq f \leq 61.5$	± 0.14	± 0.11
$61.5 < f < 63$	-	± 0.07
$63 \leq f \leq 65$	-	± 0.07
$65 < f \leq 67$	-	± 0.07

RMM-72 with 1:8 Switch

Typical

mixReceiverPath = direct

Frequency range (GHz)	Power level range (dBm)	
	$-85 \leq L_v < -60$	$-60 \leq L_v \leq +1$
$57 \leq f \leq 60.5$	± 1.36	± 1.19
$60.5 < f \leq 61.5$	± 1.83	± 1.69
$61.5 < f < 62.5$	± 2.33	± 2.13
$62.5 \leq f < 65$	± 1.68	± 1.49 ^[14]
$65 \leq f \leq 67$	± 1.31	± 0.97 ^[14]

The given values are valid after the adjustment. However, there will be a drift up to the value shown in the column *Monthly* of the next table per month over time. In case any brand new component (such as a new WSRF-6 card, a new RFIM-18, a new RMM and so on) is used, there will be an additional drift up to the value shown in the column *Initial* within the first two weeks. Afterwards, the initial drift will be gone and only the monthly drift will be taken into account.

Frequency range (GHz)	Additional Drift	
	Initial (dB)	Monthly (dB)
$57 \leq f \leq 60.5$	± 0.18	± 0.12
$60.5 < f \leq 61.5$	± 0.02	± 0.07
$61.5 < f < 62.5$	± 0.05	± 0.07
$62.5 \leq f < 65$	± 0.05	± 0.07
$65 \leq f \leq 67$	± 0.03	± 0.07

WSMM measure power

Measure power conditions:

- All values are absolute maximum ratings.

RMM-44 without switch

Absolute maximum

^[14] for $-60 \leq L_v \leq 0$ dBm

Carrier frequency range (GHz)	Power (dBm)
$24 \leq f \leq 44$	+14

RMM-44 with 1:8 switch

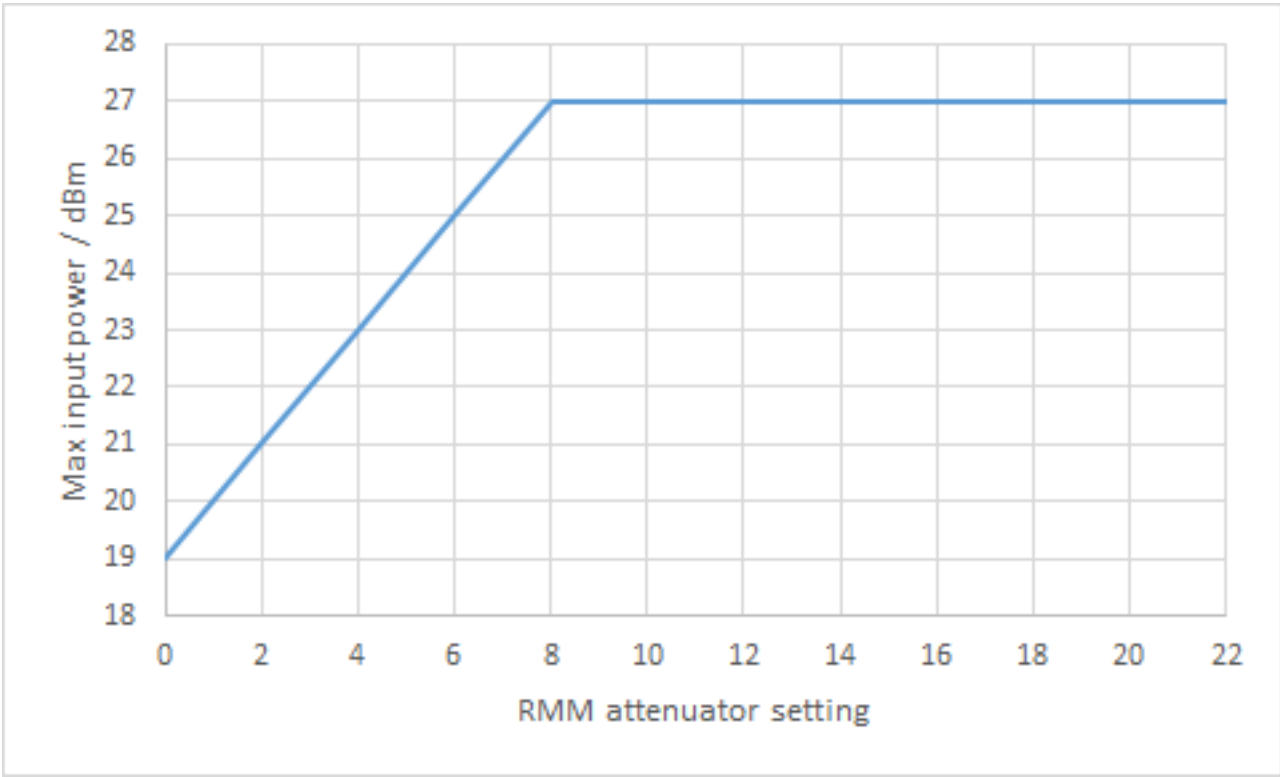
Absolute maximum

Carrier frequency range (GHz)	Power (dBm)
$24 \leq f \leq 40$	+20

RMM-53 without switch

Absolute maximum

Carrier frequency range (GHz)	Power (dBm)
$24 \leq f \leq 52.8$	+27

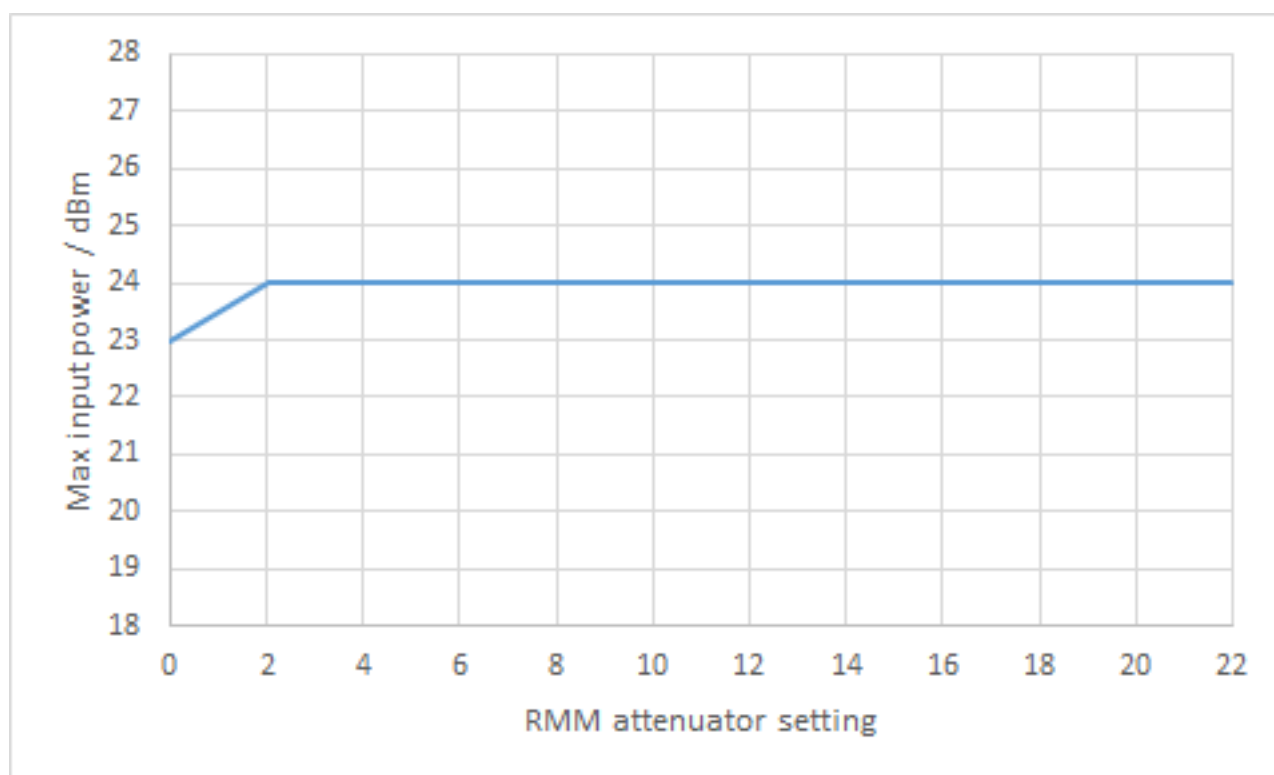


RMM53 measure power without switch

RMM-53 with 1:8 switch

Absolute maximum

Carrier frequency range (GHz)	Power (dBm)
$24 \leq f \leq 52.8$	+24



RMM53 measure power with switch

RMM-72 without switch

Absolute maximum

Carrier frequency range (GHz)	Power (dBm)
$57 \leq f < 61$	+8
$61 \leq f \leq 70$	+10

RMM-72 with 1:8 switch

Absolute maximum

Carrier frequency range (GHz)	Power (dBm)
$57 \leq f \leq 70$	+15

WSMM measure noise floor

Measure noise floor conditions and notes:

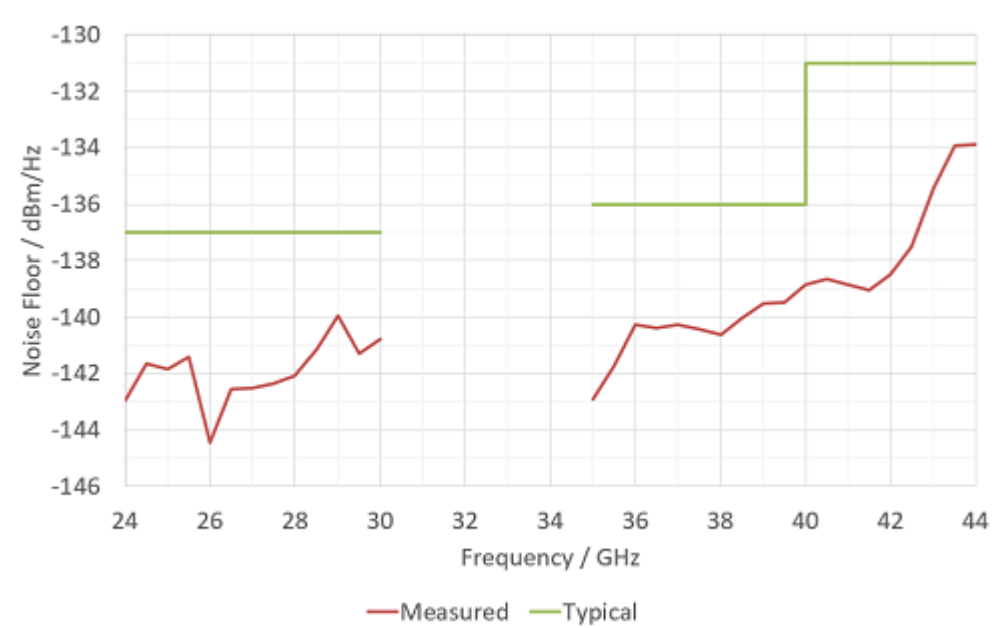
- All values are typical.
- Non-default parameters in SmarTest 8:
 - expectedMaxPower = -130 dBm
 - samples = 1024
 - sampleRate = 250 MHz
 - bandwidthOfInterest = 1 MHz
 - resultAveraging = 8
 - ifFrequency = 15 MHz

- Center frequency is programmed to the specified frequency^[15].
- Noise floor was measured with the RMM input terminated with 50 Ω.
- For optimal performance avoid generating or measuring strong signals on the neighbor channel.

RMM-44 without Switch

Typical
mixReceiverPath = direct

Frequency range (GHz)	Noise floor (dBm/Hz)
24 ≤ f ≤ 30	-137
35 ≤ f ≤ 40	-136
40 < f ≤ 44	-131



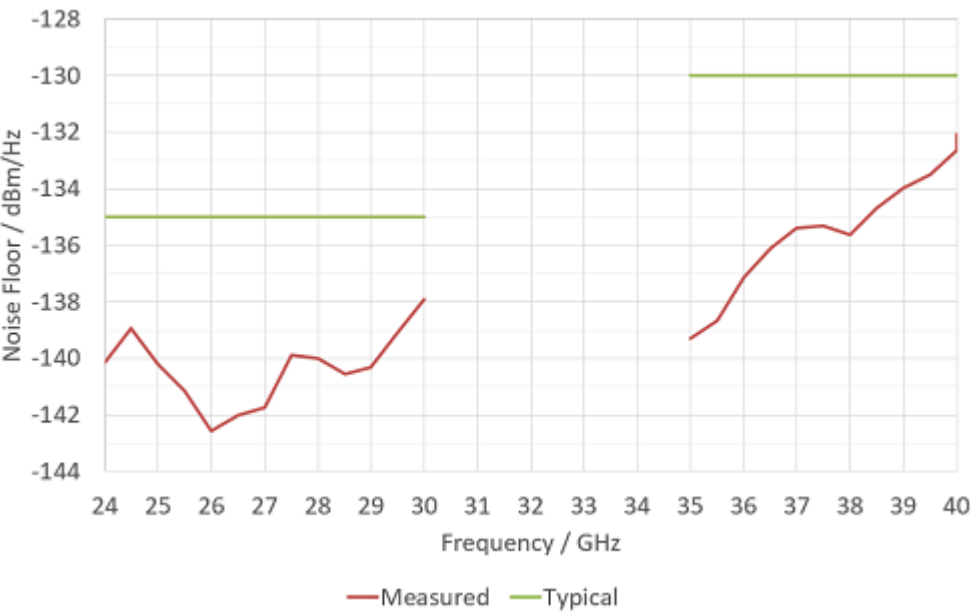
RMM-44 without switch

RMM-44 with 1:8 switch

Typical
mixReceiverPath = direct

Frequency range (GHz)	Noise floor (dBm/Hz)
24 ≤ f ≤ 30	-135
35 ≤ f ≤ 40	-130

^[15] some center frequencies were shifted to avoid spurs

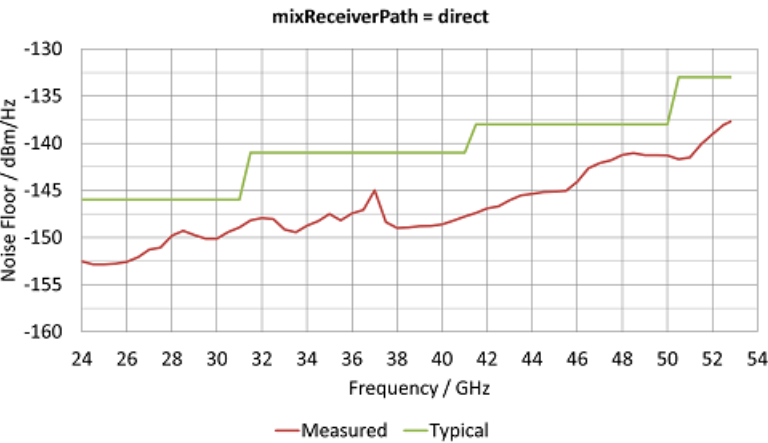


RMM-44 with 1:8 switch

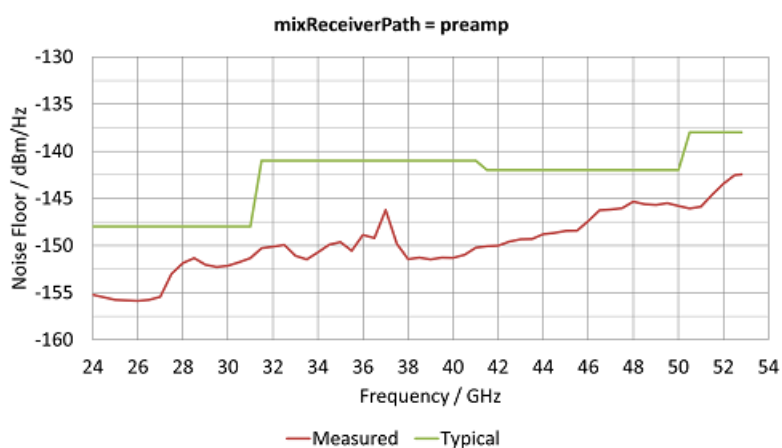
RMM-53 without switch

Typical

mixReceiverPath	Frequency range (GHz)	Noise floor (dBm/Hz)
direct	$24 \leq f \leq 31$	-146
	$31 < f \leq 41$	-141
	$41 < f \leq 50$	-138
	$50 < f \leq 52.8$	-133
preamp	$24 \leq f \leq 31$	-148
	$31 < f \leq 41$	-141
	$41 < f \leq 50$	-142
	$50 < f \leq 52.8$	-138



Meas Noise RMM53 without switch receiverPath direct

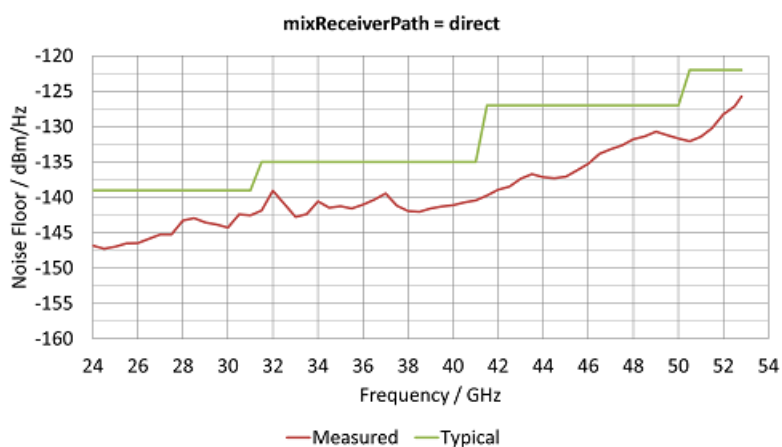


Meas Noise RMM53 without switch receiverPath preamp

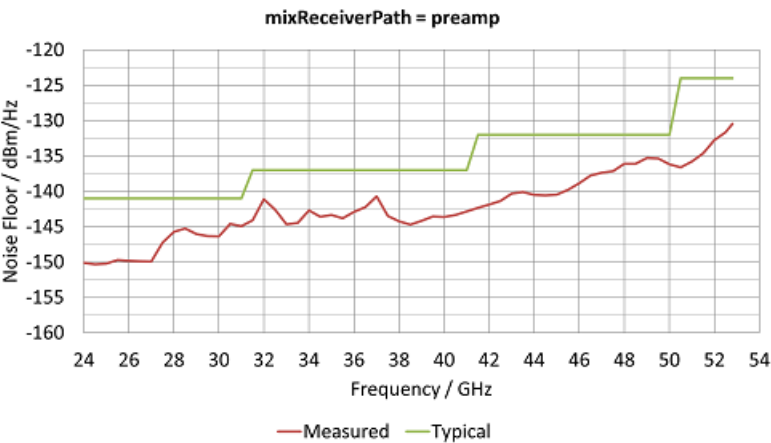
RMM-53 with 1:8 switch

Typical

mixReceiverPath	Frequency (GHz)	Noise floor (dBm/Hz)
direct	$24 \leq f \leq 31$	-139
	$31 < f \leq 41$	-135
	$41 < f \leq 50$	-127
	$50 < f \leq 52.8$	-122
preamp	$24 \leq f \leq 31$	-141
	$31 < f \leq 41$	-137
	$41 < f \leq 50$	-132
	$50 < f \leq 52.8$	-124



Meas noise RMM53 with switch receiverPath direct



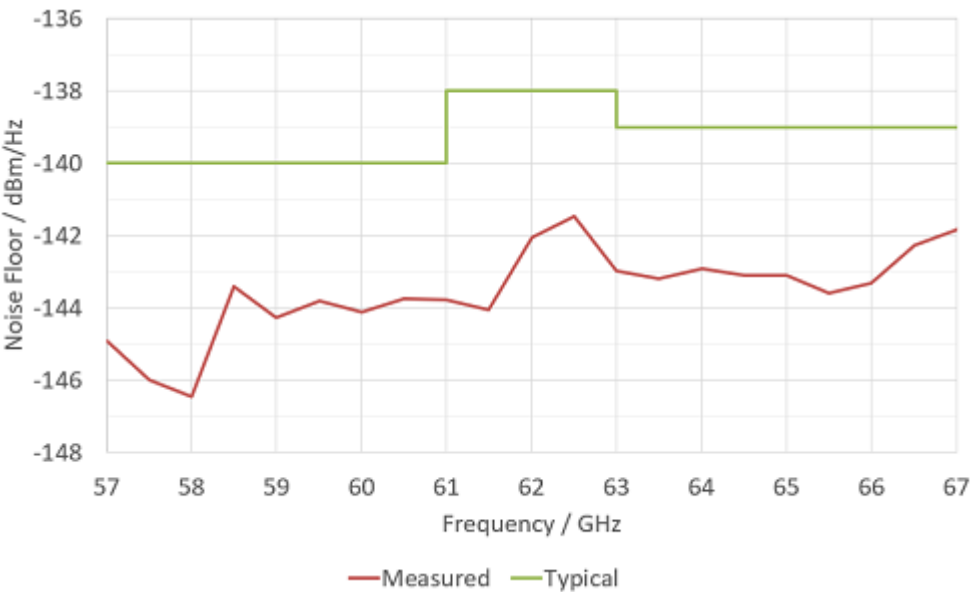
Meas noise RMM53 with switch receiverPath preamp

RMM-72 without switch

Typical

mixReceiverPath = direct

Frequency range (GHz)	Noise floor (dBm/Hz)
$57 \leq f \leq 61$	-140
$61 < f < 63$	-138
$63 \leq f \leq 67$	-139



RMM-72 without switch

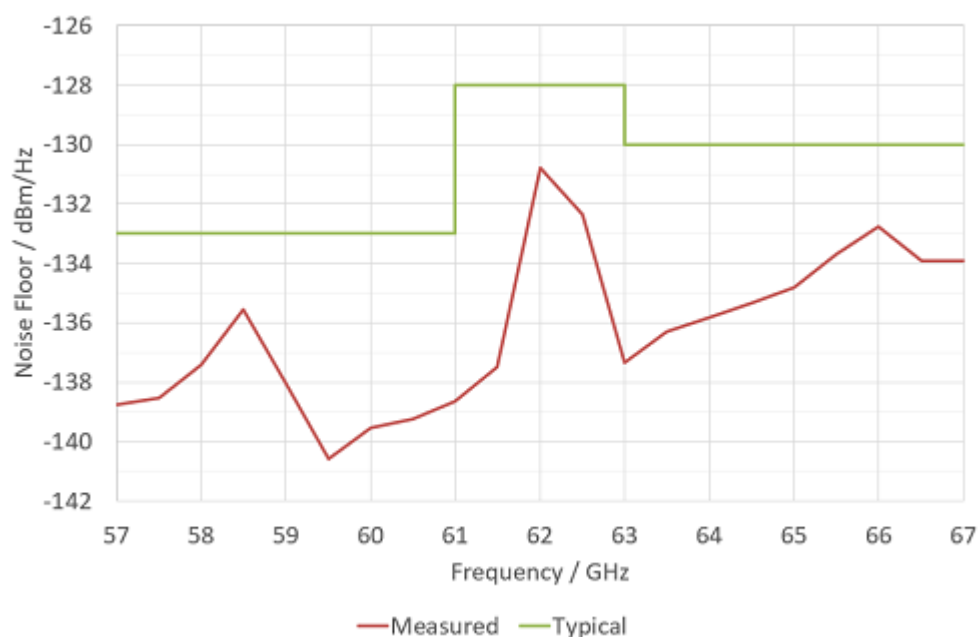
RMM-72 with 1:8 switch

Typical

mixReceiverPath = direct

Frequency range (GHz)	Noise floor (dBm/Hz)
$57 \leq f \leq 61$	-133

Frequency range (GHz)	Noise floor (dBm/Hz)
$61 < f < 63$	-128
$63 \leq f \leq 67$	-130



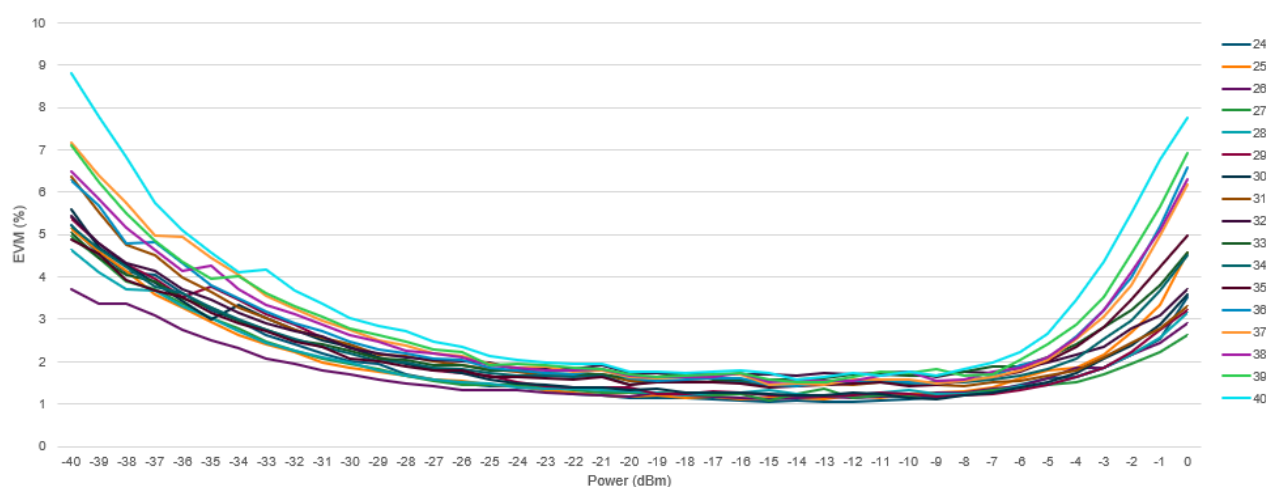
RMM-44 with 1:8 switch

WSMM measure EVM for RMM-44

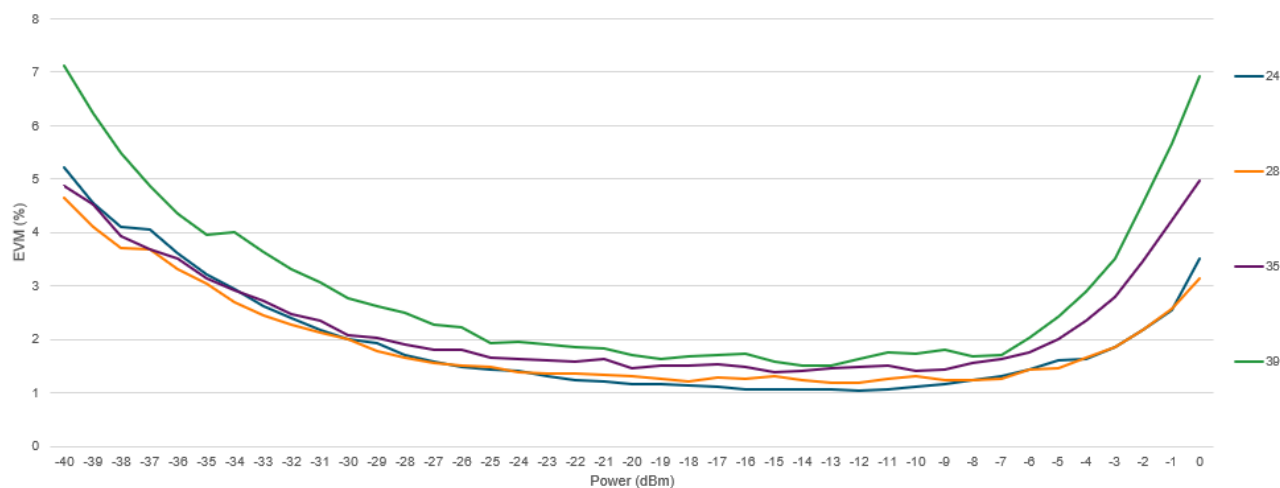
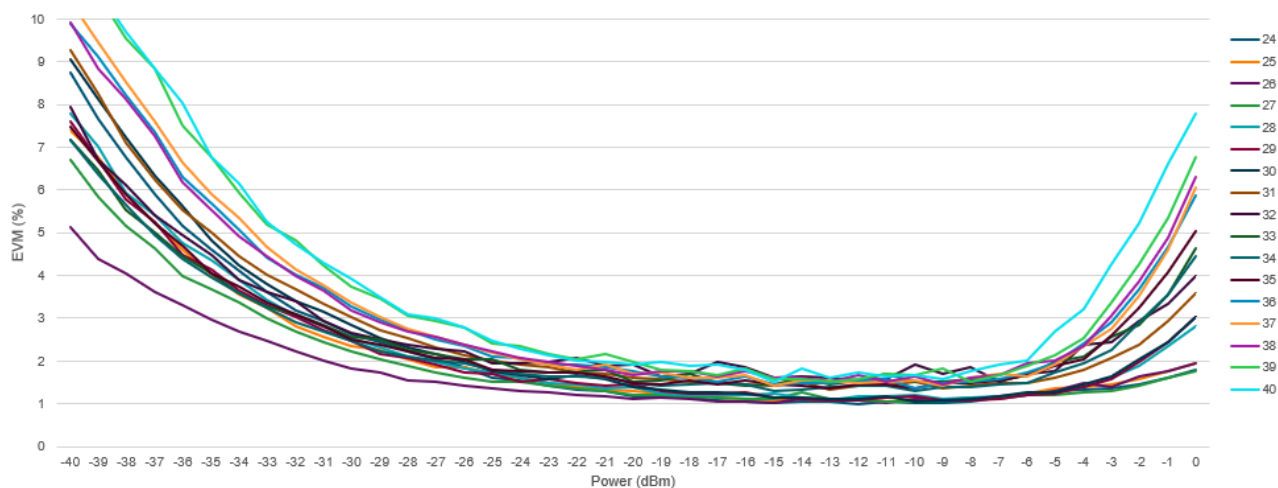
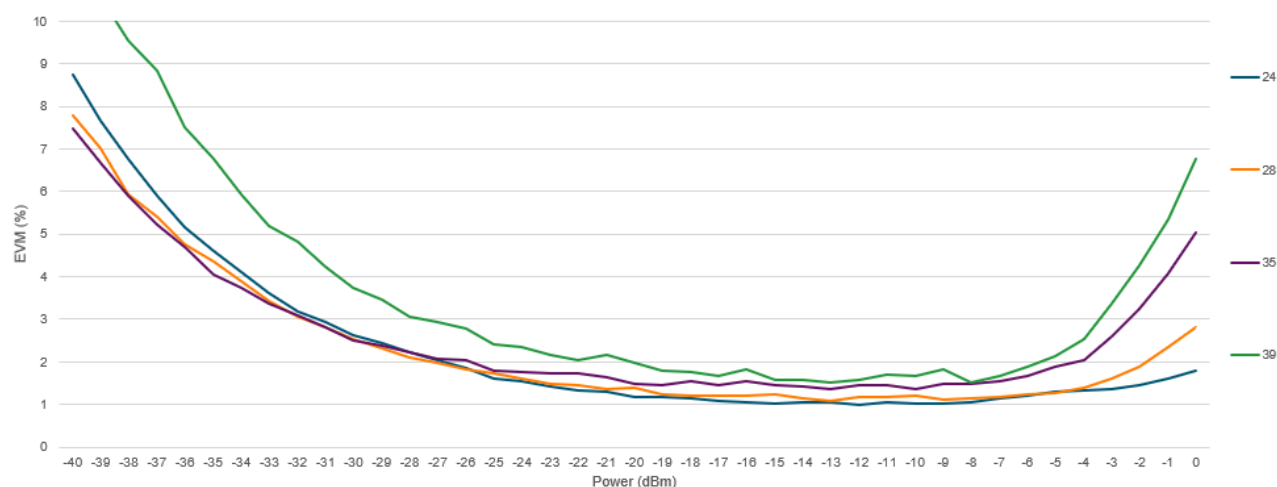
Measure EVM for RMM-44 conditions and notes:

- All values are meas values, according to *Terms and conditions* on page 4.
- Non-default parameters in SmarTest 8
 - RFIM18: Direct path
 - WSRF: default receiver path chosen by SmarTest
- Base Frequency left as default
- Waveform = NR5G_MCS27_UL_100MHz_64QAM

Measure EVM for RMM-44 without switch 24 GHz to 44 GHz



Meas EVM x1 RMM-44

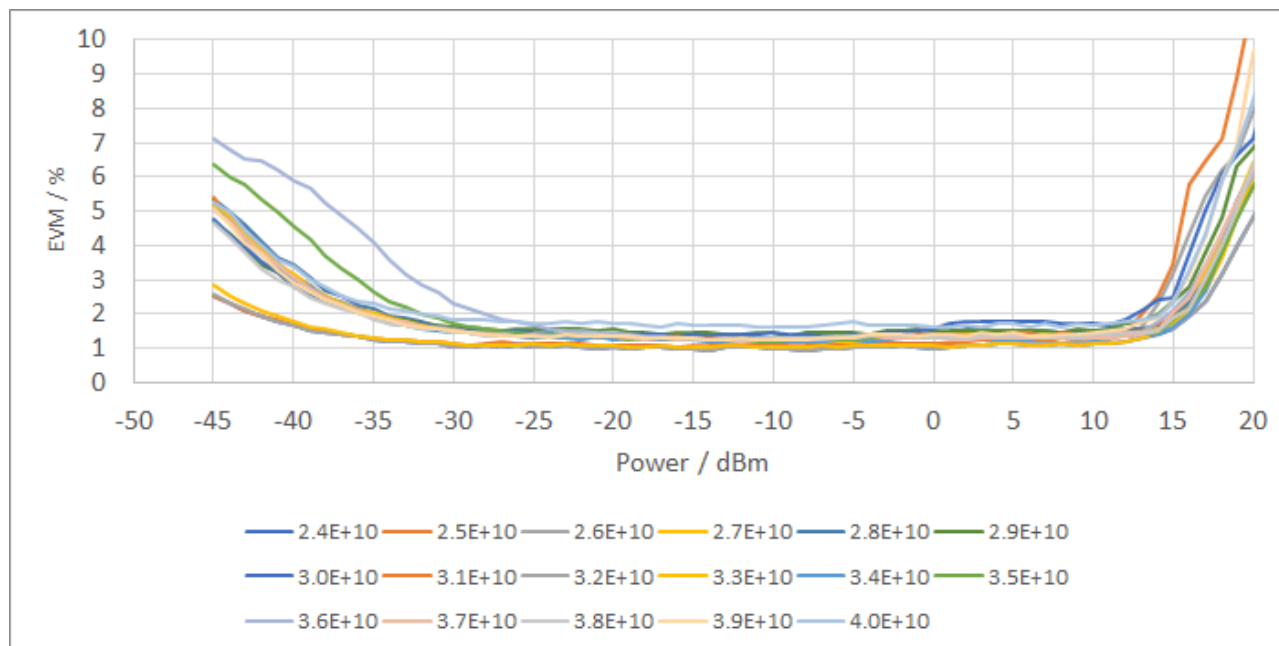
Measure EVM RMM-44 without switch 24 GHz to 44 GHz with few frequencies**Measure EVM (x1) RMM-44 without switch 24 GHz to 44 GHz with few frequencies****Measure EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz****Measure EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz****Measure EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz with few frequencies****Measure EVM RMM-44 with 1:8 switch 24 GHz to 40 GHz with few frequencies**

WSMM measure EVM for RMM-53

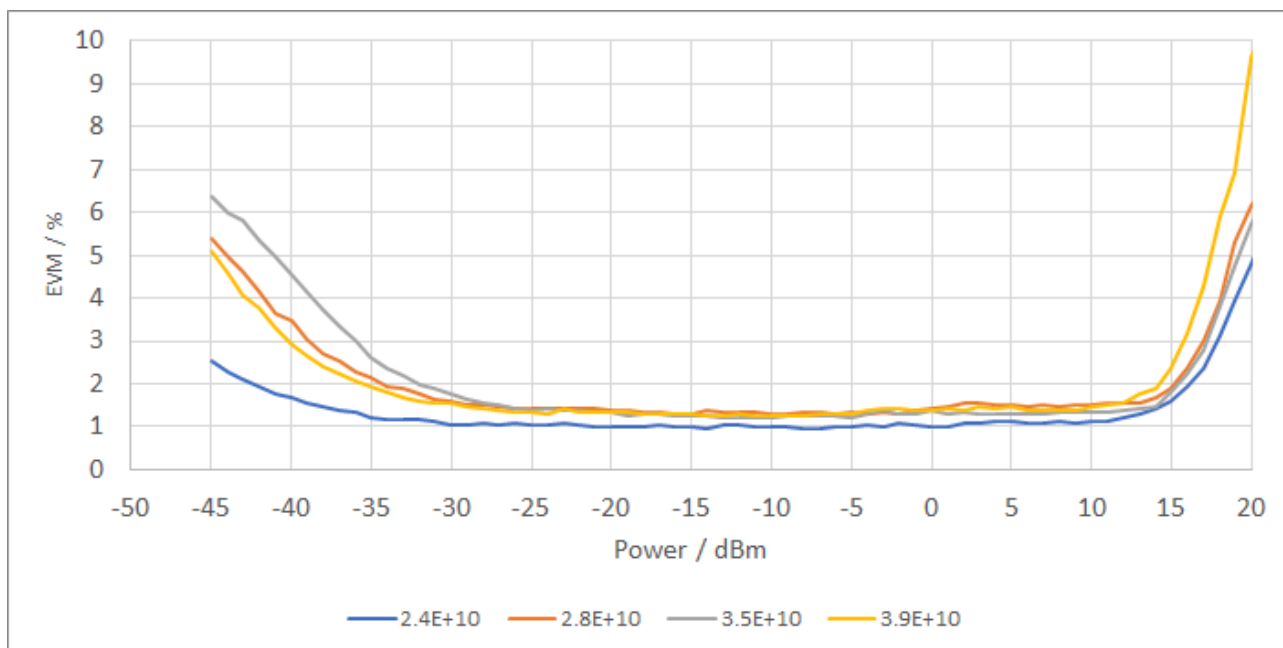
Measure EVM for RMM-53 conditions and notes:

- All values are meas values, according to *Terms and conditions* on page 4.
- Non-default parameters in SmarTest 8
 - WSRF: default receiver path chosen by SmarTest
- Base Frequency = 5.1 GHz
- Waveform = UL_MCS27_256QAM_100MHz_FR2_122p88Msps_4slots_2subfr_short
- Frequency range limited by external equipment to 40 GHz

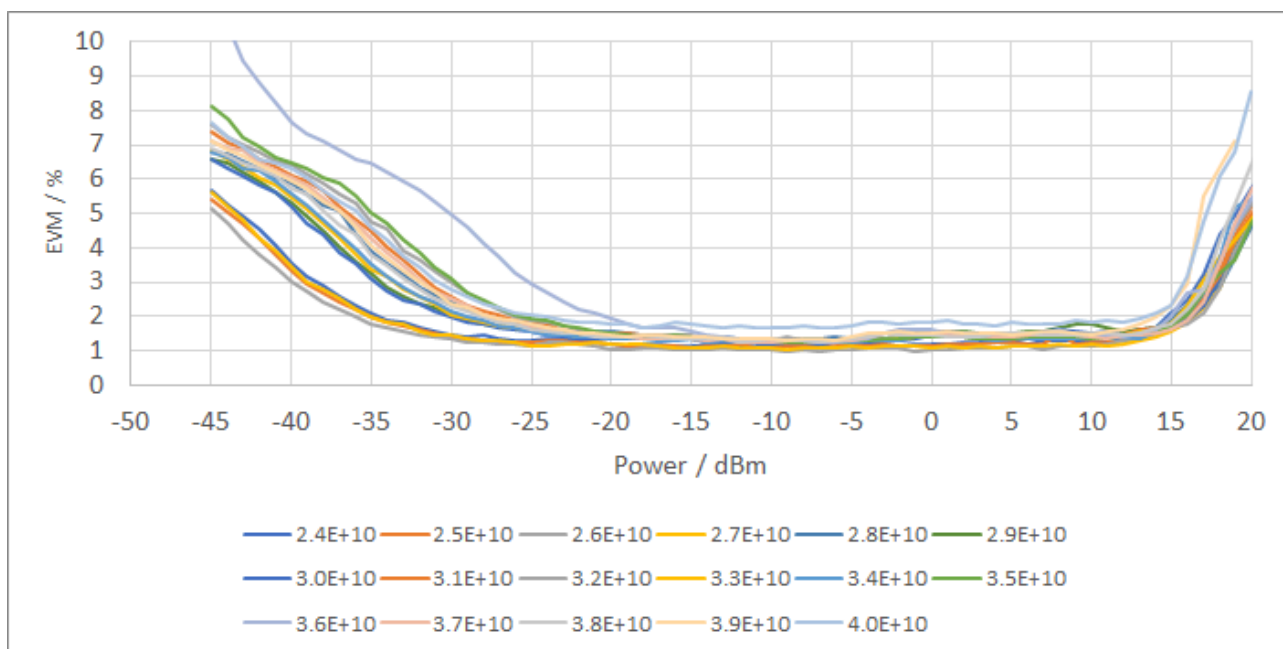
Measure EVM RMM-53 without switch 24 GHz to 40 GHz



Meas EVM RMM53 without switch

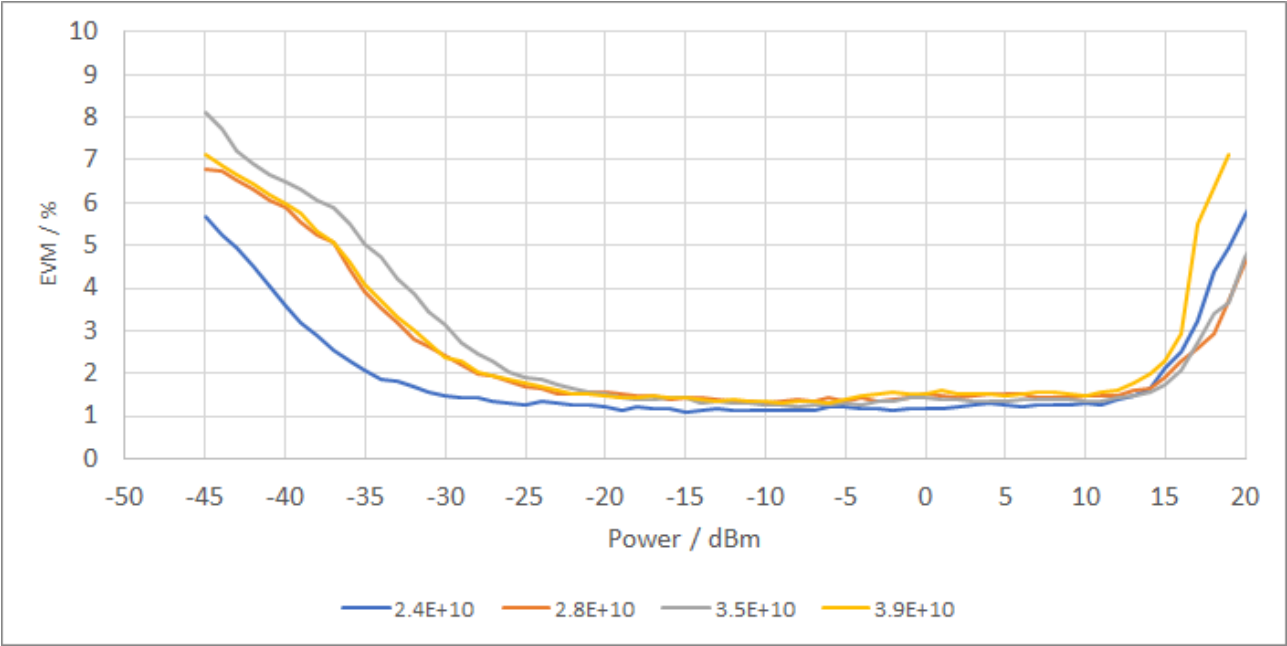
Measure EVM RMM-53 without switch with few frequencies

Meas EVM RMM53 without switch few freq

Measure EVM RMM-53 with 1:8 switch 24 GHz to 40 GHz

Meas EVM RMM53 with switch

Measure EVM RMM-53 with 1:8 switch with few frequencies



Meas EVM RMM53 with switch few freq

WSMM measure P1dB

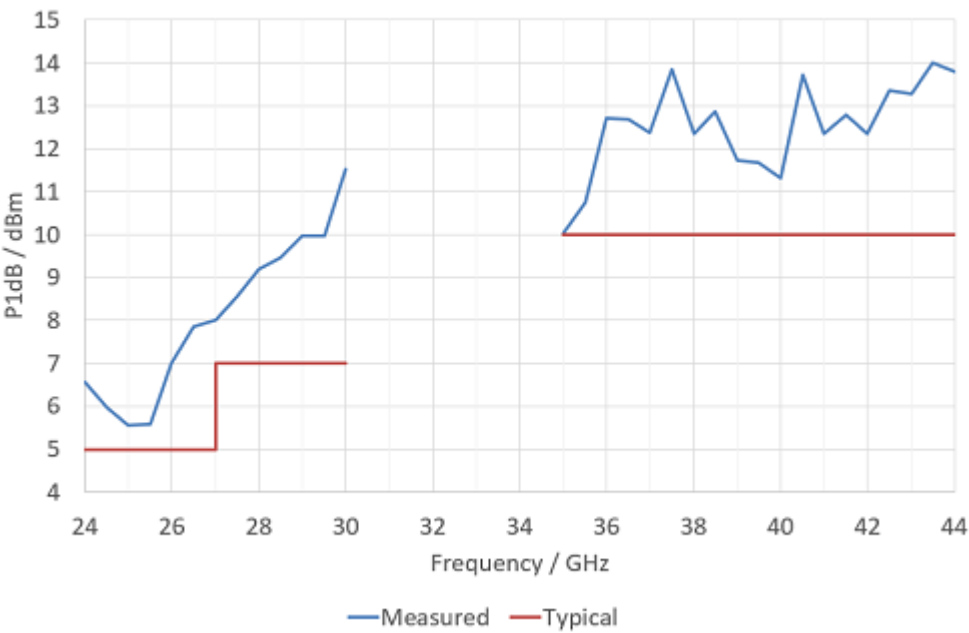
Measure P1dB conditions:

- Non-default parameters in SmarTest 8
 - mixReceiverPath = direct
 - bandwidthOfInterest = 1 kHz
 - resultAveraging = 16

RMM-44 without Switch

Typical

Carrier frequency range (GHz)	P1dB (dBm)
$24 \leq f < 27$	+5
$27 \leq f \leq 30$	+7
$35 \leq f \leq 44$	+10

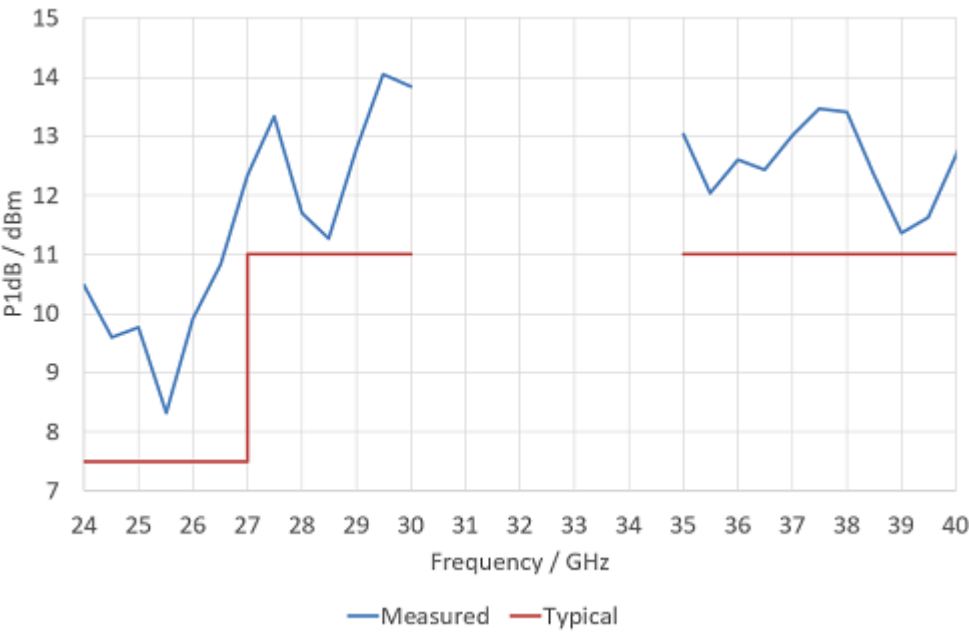


RMM-44 without switch

RMM-44 with 1:8 Switch

Typical

Carrier frequency range (GHz)	P1dB (dBm)
$24 \leq f < 27$	+7.5
$27 \leq f \leq 30$	+11
$35 \leq f \leq 40$	+11



RMM-44 with switch

RMM-53 without switch

Measured

Carrier frequency range (GHz)	P1dB (dBm)
$24 \leq f \leq 50$	+20

RMM-53 with 1:8 switch

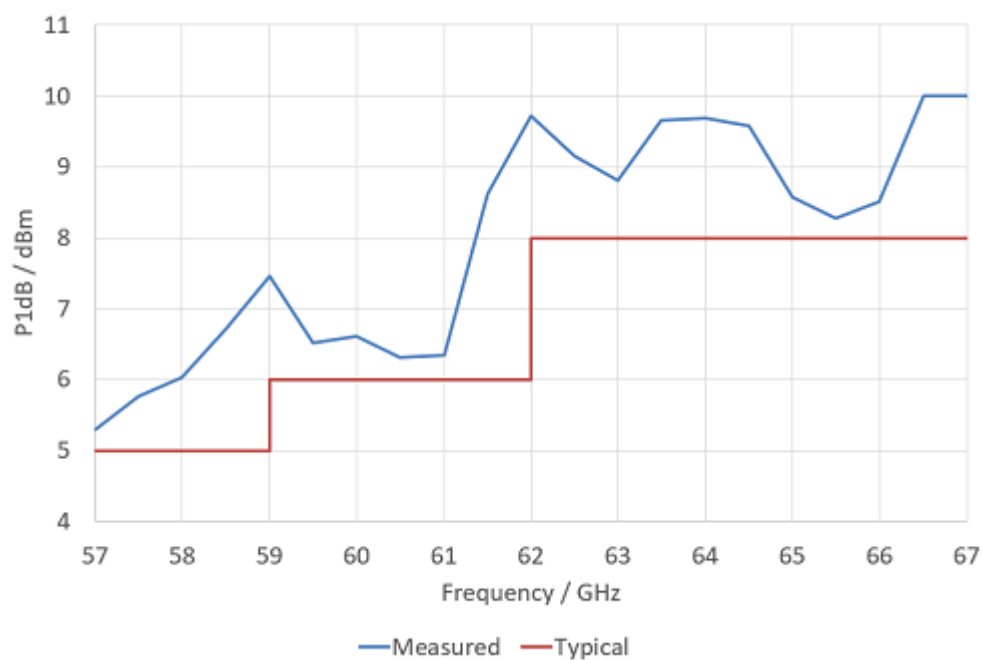
Measured

Carrier frequency range (GHz)	P1dB (dBm)
$24 \leq f \leq 48$	+24 ^[16]
$48 < f \leq 50$	+21

RMM-72 without Switch

Typical

Carrier frequency range (GHz)	P1dB (dBm)
$57 \leq f < 59$	+5
$59 \leq f < 62$	+6
$62 \leq f \leq 67$	+8



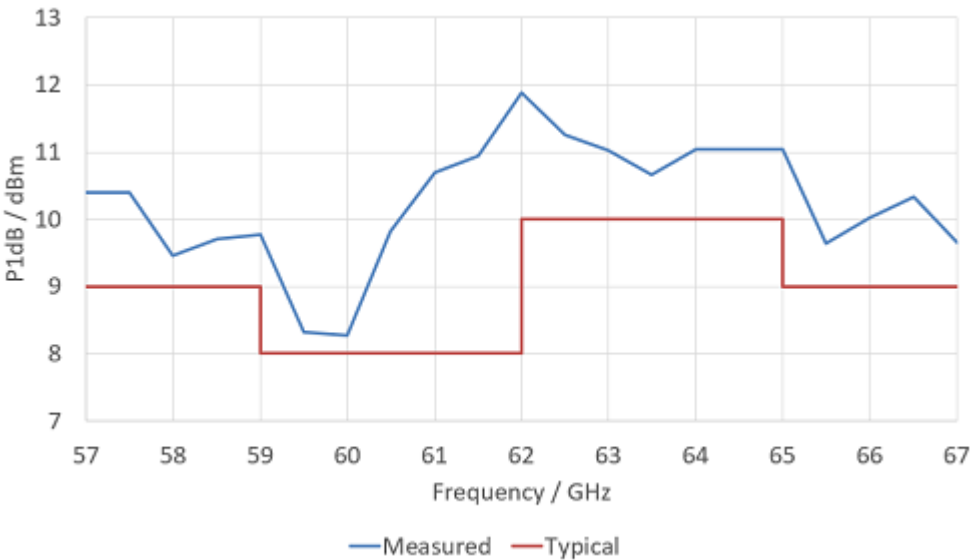
RMM-72 without switch

RMM-72 with 1:8 Switch

Typical

Carrier frequency range (GHz)	P1dB (dBm)
$57 \leq f \leq 59$	+9
$59 < f < 62$	+8
$62 \leq f \leq 65$	+10
$65 < f \leq 67$	+9

^[16] limited by abs. max. input power.



RMM-72 with switch

WSMM performance implications on other test head cards

Using the WSMM Family Board E8027FAA has performance implications on other test head cards.

Implications for specific cards:

- *DPS128* on page 43
- *PS1600* on page 43
- *PS9G* on page 44

Terms and conditions

spec	Specifications describe warranted product performance.
typ	Characteristics are included as typical values describe a performance that is not covered by the product warranty. They describe a performance that is met by 80 % of the units with a 95 % confidence level.
meas	Measured values characterize expected product performance by means of measurements provided on individual samples.

DPS128

For general technical specification see

- See topic 148671
- See topic 141967
- See topic 141968

All DPS channels have a capacitor of 2.2 nF installed. The typical trace length of DPS locations is ~25 cm, resulting in ~0.2 Ohm per trace. MS-DPS1 and MS-DPS5 locations are longer and result in ~0.4 Ohm per trace.

Ganging of DPS channels is supported on WSMM only (for example ganging of channels on DUT board).

Ganging is not supported within DPS pogo cable(s) on family board (E8027-66402) within WSMM pogo blocks.

	dl	DUT Scale interface specification			WSMM specification		
Loadstep		1 μ F	47 μ F	100 μ F	1 μ F	47 μ F	100 μ F
t _{sett} (us)	0.1 A to 1 A	20 μ s	25 μ s	30 μ s	20 μ	25 μ	30 μ
	1 A to 0.1 A	20 μ s	25 μ s	30 μ s	20 μ	25 μ	30 μ
dV (mV)	0.1 A to 1 A	320 mV	120 mV	85 mV	370 mV	125 mV	90 mV
	1 A to 0.1 A	500 mV	125 mV	85 mV	550 mV	130 mV	90 mV

PS1600

This specification is based on typical trace length of 250 mm. Approximation for shorter lengths, assumed a somewhat linear relationship, for example shortest trace 0 mm (+80 mm

cable which is ~20 mm trace) which is about 10% expected specification degradation: impact for 0 mm trace length is assumed to be ~10% of longest trace impact.

	Impacting or boundary condition	PS1600 DUT Scale interface specification	PS1600 WSMM specification (typical)
DC trace wiring resistance	Voltage accuracy (force and measure)	0.5 Ohm	1...2 Ohm 0...251 mm trace length
Driver, AC Max transition times	swing 1.0 V / 1.8 V / 3.0 V	0.6 ns / 0.7 ns / 0.8 ns	0.67 ns / 0.78 ns / 0.9 ns
Driver, AC min pulse width	swing 1.0 V / 1.8 V / 3.0 V	0.7 ns / 0.8 ns / 0.9 ns	0.78 ns / 0.9 ns / 1.0 ns
Receiver, AC Min detectable pulse width		600 ps	620 ps
Receiver, AC Intrinsic transition time		350 ps	450 ps
OTA and EPA, system		< ± 200 ps (1600 Mbps) EPA = OTA / 2	< ± 220 ps (1600 Mbps) < ± 215 ps (1400 Mbps) EPA = OTA / 2
OTA and EPA, per-board		< ± 160 ps (1600 Mbps) EPA = OTA / 2	< ± 180 ps (1600 Mbps) < ± 175 ps (1600 Mbps) EPA = OTA / 2

PS9G

This specification is based on a typical trace length of 445 mm. Approximation for shorter lengths, assumed a somewhat linear relationship, for example, shortest trace 364 mm which is about 70% expected specification degradation: impact for 364 mm trace length is assumed to be ~70% of longest trace impact.

PS9G WSMM specification (typical), typical trace, 445 mm narrow

	Impacting or boundary condition	PS9G DUT Scale interface specification (typical)	PS9G WSMM specification (typical), typical trace, 445 mm narrow
DC trace wiring resistance	Voltage accuracy (force and measure)	0.5 Ohm	2...3 Ohm 364...557 mm trace length
Eye Height, single ended Driver (typ)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 89 % of Vswing 5.33 Gbps 70 % of Vswing 6.8 Gbps 70 % of Vswing	2.5 Gbps 70 % of Vswing 5.33 Gbps 40 % of Vswing 6.8 Gbps 35 % of Vswing

	Impacting or boundary condition	PS9G DUT Scale interface specification (typical)	PS9G WSMM specification (typical), typical trace, 445 mm narrow
Eye Height, differential Driver (typ)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 85 % of Vswing 5.33 Gbps 70 % of Vswing 8 Gbps 70 % of Vswing	2.5 Gbps 65 % of Vswing 5.33 Gbps 40 % of Vswing 8 Gbps 36 % of Vswing
Eye Height, single ended Driver + Receiver	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 88 % of Vswing 5.33 Gbps 68 % of Vswing 6.8 Gbps 54 % of Vswing	2.5 Gbps 47 % of Vswing 5.33 Gbps 10 % of Vswing
Eye Height, differential Driver + Receiver (typ)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 80 % of Vswing 5.33 Gbps 68 % of Vswing 8 Gbps 35 % of Vswing	2.5 Gbps 39 % of Vswing 5.33 Gbps 3 % of Vswing
Eye Height, single ended Driver + Receiver (min)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 78 % of Vswing 5.33 Gbps 29 % of Vswing 6.8 Gbps 15 % of Vswing	2.5 Gbps 37 % of Vswing
Eye Height, differential Driver + Receiver (min)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 74 % of Vswing 5.33 Gbps 37 % of Vswing 8 Gbps 11 % of Vswing	2.5 Gbps 33 % of Vswing
OTA at 4.5 Gbps Longest trace length	See "PS9G OTA/EPA (Topic 127348)", 4.5 Gbps per pin pair / per-board	< ± 35 ps / < ± 46 ps	< ± 88 ps / < ± 99 ps
OTA at 2.5 Gbps Longest trace length	See "PS9G OTA/EPA (Topic 127348)", 2.5 Gbps per pin pair / per-board	Not specified	< ± 55 ps / < ± 66 ps

PS9G WSMM specification (typical), typical trace, 120 mm narrow

	Impacting or boundary condition	PS9G DUT Scale interface specification (typical)	PS9G WSMM specification (typical), typical trace, 120 mm narrow
DC trace wiring resistance	Voltage accuracy (force and measure)	0.5 Ohm	2..3 Ohm 364...557 mm trace length
Eye Height, single ended Driver (typ)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 89 % of Vswing 5.33 Gbps 70 % of Vswing 6.8 Gbps 70 % of Vswing	2.5 Gbps 84 % of Vswing 5.33 Gbps 65 % of Vswing 6.8 Gbps 62 % of Vswing
Eye Height, differential Driver (typ)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 85 % of Vswing 5.33 Gbps 70 % of Vswing 8 Gbps 70 % of Vswing	2.5 Gbps 80 % of Vswing 5.33 Gbps 65 % of Vswing 8 Gbps 61 % of Vswing
Eye Height, single ended Driver + Receiver	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 88 % of Vswing 5.33 Gbps 68 % of Vswing 6.8 Gbps 54 % of Vswing	2.5 Gbps 79 % of Vswing 5.33 Gbps 53 % of Vswing 8 Gbps 35 % of Vswing
Eye Height, differential Driver + Receiver (typ)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 80 % of Vswing 5.33 Gbps 68 % of Vswing 8 Gbps 35 % of Vswing	2.5 Gbps 70 % of Vswing 5.33 Gbps 46% of Vswing 8 Gbps 16 % of Vswing
Eye Height, single ended Driver + Receiver (min)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 78 % of Vswing 5.33 Gbps 29 % of Vswing 6.8 Gbps 15 % of Vswing	2.5 Gbps 68 % of Vswing 5.33 Gbps 15% of Vswing 8 Gbps 10 % of Vswing
Eye Height, differential Driver + Receiver (min)	See "PS9G Eye Height (Topic 127350)"	2.5 Gbps 74 % of Vswing 5.33 Gbps 37 % of Vswing 8 Gbps 11 % of Vswing	2.5 Gbps 66 % of Vswing 5.33 Gbps 26% of Vswing
OTA at 4.5 Gbps Longest trace length	See "PS9G OTA/EPA (Topic 127348)", 4.5 Gbps per pin pair / per-board	< ± 35 ps / < ± 46 ps	< ± 88 ps / < ± 99 ps

	Impacting or boundary condition	PS9G DUT Scale interface specification (typical)	PS9G WSMM specification (typical), typical trace, 120 mm narrow
OTA at 2.5 Gbps Longest trace length	See "PS9G OTA/EPA (Topic 127348)", 2.5 Gbps per pin pair / per-board	Not specified	$< \pm 55 \text{ ps} / < \pm 66 \text{ ps}$

PS9G WSMM specification (typical), typical trace, 300 mm wide

	Impacting or boundary condition	PS9G DUT Scale interface specification (typical)	PS9G WSMM specification (typical), typical trace, 300 mm wide
OTA at 4.5 Gbps	See "PS9G OTA/EPA (Topic 127348)", 4.5 Gbps per pin pair / per-board	$< \pm 35 \text{ ps} / < \pm 46 \text{ ps}$	$< \pm 41 \text{ ps} / < \pm 52 \text{ ps}$

PS9G WSMM specification (typical), typical trace, 250 mm wide

	Impacting or boundary condition	PS9G DUT Scale interface specification (typical)	PS9G WSMM specification (typical), typical trace, 250 mm narrow
OTA at 4.5 Gbps	See "PS9G OTA/EPA (Topic 127348)", 4.5 Gbps per pin pair / per-board	$< \pm 35 \text{ ps} / < \pm 46 \text{ ps}$	$< \pm 45 \text{ ps} / < \pm 56 \text{ ps}$

Assuming ~40% eye height the

- pogo block 301 (16 channels) can support 3 Gbps and the
- pogo blocks 302, 305, 306 (48 channels) can support 6 Gbps.